Atiyah-Macdonald Exercises

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1 Rings and Ideals (Complete)

1.1 Exercise

Proof. By assumption, $x^m = 0$ for some m. Then,

$$(1+x)\sum_{i=0}^{m-1}(-1)^i x^i = 1 + (-1)^{m-1}x^m = 1,$$

so $1+x$ is a unit.

Let x be nilpotent and u be a unit. Then, $u + x = u(1 + u^{-1}x)$. But $u^{-1}x$ is nilpotent since x is. By the previous result, $1 + u^{-1}x$ is a unit and hence $u(1 + u^{-1}x) = u + x$ is also a unit. □

1.2 Exercise

(i) Proof. We will prove this by induction. Suppose f has degree 0 with inverse $g = b_0 + \cdots + b_m x^m$. Then, $1 = fb_0 + \cdots + fb_m x^m$. Therefore, since $f \in A$ and $fb_0 = 1$, f is a unit. The converse is trivial since $f = a_0$, proving the base case.

Now, suppose that for all polynomials of degree $n-1$, the proposition holds. Suppose f has degree n and has an inverse $g = b_0 + \cdots + b_m x^m$. Then,

$$\begin{aligned} 1 &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} b_j x^j\right) \\ &= \sum_{k=0}^{n+m}\left(\sum_{i=0}^{k} a_i b_{k-i}\right) x^i. \end{aligned}$$

This implies $a_n b_m = 0$. We'll show by induction on r that $a_n^{r+1} b_{m-r} = 0$, the base case already proven. We assume this is true for all $r < l$. Multiplying by a_n^l,

$$\begin{aligned} a_n^l &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} a_n^l b_j x^j\right) \\ &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m-l} a_n^l b_j x^j\right), \end{aligned}$$

where we used the inductive hypothesis to remove higher terms: if $j > m-l$, then $l-(m-j+1) \geq 0$, so $a_n^l b_j = a_n^{l-(m-j+1)}(a_n^{m-j+1} b_{m-(m-j)}) = 0$. The coefficient of highest degree is $a_n \cdot a_n^l b_{m-l}$, so $a_n^{l+1} b_{m-l} = 0$. By induction, $a_n^{r+1} b_{m-r} = 0$ for all r.

From the expansion of $f \cdot g = 1$, we know that $a_0 b_0 = 1$. Then, using our result,

$$a_n^{m+1} = a_0(a_n^{m+1} b_{m-m}) = a_0 \cdot 0 = 0,$$

so a_n is nilpotent. Then, $-a_n x^n$ is nilpotent. Since f is a unit, by Exercise 1.1, $f + (-a_n x^n)$ is also a unit. However, it has degree $n-1$. By the inductive hypothesis, a_0 is a unit and a_i is nilpotent for $1 \leq i < n$. We've shown a_n is nilpotent, so this direction is complete.

Conversely, suppose a_0 is a unit and that a_i is nilpotent for $1 \leq i \leq n$. Then, $f - a_n x^n$ has degree $n-1$, so by the inductive hypothesis, $f - a_n x^n$ is a unit since its coefficients have the right properties. However, $a_n x^n$ is nilpotent since a_n is, so by Exercise 1.1, $(f - a_n x^n) + a_n x^n = f$ is a unit, so both directions are complete. By induction, the proposition follows for f of any degree. □

(ii) Proof. We'll prove this by induction on the degree. The base case is trivial since $f = a_0$ in degree 0. Now, suppose for all degrees less than n, the proposition holds.

If f has degree n and is nilpotent, there is an m such that $f^m = 0$. Expanding, the top term is $a_n^m x^{nm}$, which is zero. Therefore, a_n is nilpotent. However, $g = f + (-a_n x^n)$ is also nilpotent since it is the sum of two nilpotent elements (if a and b are the required nilpotent powers, take $a + b$; this power will annihilate all terms in the binomial expansion of g). g has degree at most $n - 1$, so by the inductive hypothesis $a_0, \ldots, a_{n-1}$ are all nilpotent. Since a_n is nilpotent, this direction is complete.

Conversely, if f has degree n and all coefficients are nilpotent, take $f = g + a_n x^n$, where g has degree at most $n - 1$. By the inductive hypothesis, g is nilpotent, so f is the sum of two nilpotent elements. Thus, f is nilpotent. By induction, the proposition follows for f of any degree. □

(iii) Proof. The reverse implication is by definition. For the forward implication, suppose f is a zero-divisor. The set of all possible g such that $fg = 0$ has minimal degree m since it is non-empty. Choose g of degree m such that $fg = 0$. If the highest terms are $a_n x^n$ and $b_m x^m$, then $a_n b_m x^{n+m} = 0$ by $fg = 0$, so $a_n b_m = 0$. Note that $a_n g$ annihilates the term of degree m and annihilates f, so $a_n g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $a_{n-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $a_{n-r} g = 0$ for all $r < k$. Then,

$$0 = g \sum_{i=0}^{n} a_i x^i = \sum_{i=0}^{n} (g a_i) x^i = \sum_{i=0}^{n-k} (g a_i) x^i.$$

The top term is $b_m a_{n-k} x^{m+n-k}$, so $b_m a_{n-k} = 0$. Thus, $a_{n-k} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $a_{n-k} g = 0$. By induction, $a_{n-r} g = 0$ for all r.

For any a_i, we have $a_i g = 0$. Thus,

$$0 = a_i \left(b_0 + \cdots + b_m x^m\right) = a_i b_0 + \cdots + a_i b_m x^m.$$

By definition, $b_m \neq 0$, yet $a_i b_m x^m = 0$. So, $a_i b_m = 0$ for all i. Thus,

$$b_m f = b_m \sum_{i=0}^{n} (b_m a_i) x^i = 0,$$

proving the claim. □

Remark. *For use in Exercise 1.3, we just proved: if $fg = 0$, g annihilates every coefficient of f.*

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x] \to (A/\mathfrak{m})[x]$, $0 = \overline{fg} = \overline{f}\,\overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then,

$$fg = \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k.$$

Obviously, $a_i b_{k-i} \in \mathfrak{m}$ since $a_i \in \mathfrak{m}$, so any finite sum of these terms is also. Therefore, the coefficients of fg all lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.3 Exercise

Here, we will frequently use the fact that

$$A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n],$$

which is easy to see by simply grouping terms on the left-hand side by x_n-degree.

(ii) Proof. We'll prove this by induction, using Exercise 1.2(ii), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Let $f \in A[x_1, \ldots, x_n]$. We can write $f = f_0 + \cdots + f_m x_n^m \in A[x_1, \ldots, x_{n-1}][x_n]$. Assume f is nilpotent. By Exercise 1.2(ii), each f_i is nilpotent. Then, by the inductive hypothesis for (ii), all coefficients are nilpotent for each f_i and thus for f. Conversely, if all coefficients of f are nilpotent, then all coefficients of each f_i are nilpotent. By the inductive hypothesis for (ii), each f_i is nilpotent. By Exercise 1.2(ii), f is nilpotent. By induction, the result follows. □

(i) Proof. We'll prove this by induction, using Exercise 1.2(i), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Suppose f is a unit in $A[x_1, \ldots, x_n] = A[x_1, \ldots, x_{n-1}][x_n]$. Then, $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, where $f_i \in A[x_1, \ldots, x_{n-1}]$. By the inductive hypothesis for (i), f_0 is a unit and f_i is nilpotent for $i > 0$. By the inductive hypothesis for (ii), all the coefficients of f_i are nilpotent for $i > 0$. By the inductive hypothesis for f_0, the constant coefficient is a unit, and all other coefficients are nilpotent. Thus, viewing f in $A[x_1, \ldots, x_n]$, the constant term is a unit and all other coefficients are nilpotent.

Conversely, suppose f in $A[x_1, \ldots, x_n]$ with the constant term a unit and all other coefficients nilpotent. Then, we can group terms so that $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, viewing $f \in A[x_1, \ldots, x_{n-1}][x_n]$. Then, f_0 has a unit for a constant term, with all other coefficients being nilpotent. Thus, f_0 is a unit by the inductive hypothesis for (i). Likewise, f_i for $i > 0$ has all nilpotent coefficients, so Exercise 1.3(ii) shows that f_i is nilpotent. By Exercise 1.2(i), f is a unit. By induction, (i) is proven in the general case. □

(iii) Proof. First, we start with a lemma: we want to show by induction that for $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$, we have that g annihilates every coefficient of f. The base case is shown in Exercise 1.2(iii).

Suppose f is a zero-divisor and $f \in A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n]$. The set of all possible g such that $fg = 0$ has minimal degree m in x_n since it is non-empty. Choose g of degree m in x_n such that $fg = 0$. If the highest terms are $f_k x_n^k$ and $g_l x_n^l$, then $f_k g_l x^{k+l} = 0$ by $fg = 0$, so $f_k g_l = 0$. Note that $f_k g$ annihilates the term of degree m and annihilates f, so $f_k g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $f_{k-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $f_{k-r} g = 0$ for all $r < q$. Then,

$$0 = g \sum_{i=0}^{k} f_i x_n^i = \sum_{i=0}^{k} (g f_i) x_n^i = \sum_{i=0}^{k-q} (g f_i) x_n^i.$$

The top term is $g_l f_{k-q} x_n^{l+k-q}$, so $g_l f_{k-q} = 0$. Thus, $f_{k-q} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $f_{k-q} g = 0$. By induction, $f_{k-r} g = 0$ for all r.

For any f_i, we have $f_i g = 0$. Thus,

$$0 = f_i \left(g_0 + \cdots + g_m x_n^m\right) = f_i g_0 + \cdots + f_i g_m x_n^m.$$

For any g_j, we have $f_i g_j x_n^j = 0$. So, $f_i g_j = 0$ for all i and j. By induction, since $f_i, g_j \in A[x_1, \ldots, x_{n_1}]$, g_j annihilates every coefficient of f_i. Since this is true for all j, g annihilates every coefficient of f_i for all i. By induction, for all n, if $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then g annihilates every coefficient of f individually. This proves the lemma.

With this lemma, we can prove (iii). The reverse implication is by definition. For the forward implication, we prove (iii) by induction, using the base case established by Exercise 1.2(iii). Assume the inductive hypothesis up to $n-1$ and that $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$. By our lemma, and since f and g are symmetrically defined, f annihilates every coefficient of g individually; since g is nonzero, it has some nonzero coefficient $g_\beta \in A$. Thus, $g_\beta f = 0$. By induction, if for any n, $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then $af = 0$ for some nonzero $a \in A$. □

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$, $0 = \overline{fg} = \overline{f} \cdot \overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then, consider the same map as before, $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$. Then, $\overline{fg} = \overline{f} \cdot \overline{g} = 0 \cdot \overline{g} = 0$. In particular, all the coefficients of fg lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.4 Exercise

Proof. Since all maximal ideals are prime ideals, the nilradical (intersection of all prime ideals) is contained in the Jacobson radical (intersection of all maximal ideals) for any ring. Thus, we need to show that any $f(x)$ in the Jacobson radical is nilpotent.

By Proposition 1.9, if $f(x)$ is in the Jacobson radical, then $1 + xf(x)$ is a unit in $A[x]$. We can write this as

$$1 + xf(x) = 1 + a_0 x + \cdots + a_n x^{n+1}.$$

By Exercise 1.2(i), $a_0, \ldots, a_n$ are nilpotent. Thus, by Exercise 1.2(ii), f is nilpotent. □

1.5 Exercise

(i) Proof. Suppose $f = \sum_i a_i x^i \in A[[x]]$ is a unit so that there is some $g = \sum_i b_i x^i \in A[[x]]$ yielding $fg = 1$. Expanding, the constant term is $a_0 b_0 = 1$. Thus, a_0 is a unit in A.

We will prove the converse by induction; we will show that for all n, we can construct some $g_n \in A[x]$ (realized as a subring of $A[[x]]$) of degree $n-1$ such that $fg_n - 1 \in (x^n)$ and $g_n \equiv g_r \mod (x^r)$ for $r < n$. Suppose $f = \sum_i a_i x^i$ and a_0 is a unit so that there is a b_0 yielding $a_0 b_0 = 1$. Let $g_0 = b_0$. Then,

$$fg_0 - 1 = \sum_{i=0}^{\infty} a_i b_0 x^i - 1 = \sum_{i=1}^{\infty} a_i b_0 x^i + (a_0 b_0 - 1) = \sum_{i=1}^{\infty} a_i b_0 x^i \in (x^1),$$

the ideal generated by x. This proves the base case. Our inductive hypothesis is that there is some $g_{n-1} = \sum_{i=0}^{n-1} b_i x^i$ such that $fg_{n-1} - 1 \equiv 0 \mod (x^n)$. Define $b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}$, recalling that

a_0 is a unit. We claim that $g_n = \sum_{i=0}^{n} b_i x^i$ satisfies the proper congruence. We have

$$\begin{aligned}
fg_n &= \sum_{k=0}^{\infty} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k \\
&= 1 + \left(\sum_{i=0}^{n} a_i b_{n-i} \right) x^n + \sum_{k=n+1}^{\infty} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k \\
&\equiv 1 + \left(a_0 b_n + \sum_{i=1}^{n} a_i b_{n-i} \right) x^n \mod (x^{n+1}) \\
&\equiv 1 + \left(a_0 \left(-a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i} \right) + \sum_{i=1}^{n} a_i b_{n-i} \right) x^n \mod (x^{n+1}) \\
&\equiv 1 + 0 \cdot x^n \mod (x^{n+1}) \\
&\equiv 1 \mod (x^{n+1}).
\end{aligned}$$

Note that since only the x^n coefficient was changed, the compatibility requirement holds: i.e., $g_n \equiv g_r \mod (x^r)$ for $r < n$. Thus, if we let $g = \sum_{i=0}^{\infty} b_i x^i$, then $fg - 1 \equiv 0 \mod (x^n)$ for all n. Once we show that $\bigcap_n (x^n) = (0)$, then we'll have $fg = 1$, proving the claim. But this is clear since any nonzero element $h \in A[[x]]$ has some nonzero coefficient h_j, so $h \notin (x^{j+1})$. □

(ii) Proof. Suppose $f = \sum_i a_i x^i$ is nilpotent. We will show that each a_i is nilpotent by induction. For the base case, note that if $f^m = 0$, then the constant term is $a_0^m = 0$, so $f_0 = a_0$ is nilpotent. For the inductive step, suppose that a_i is nilpotent for $i < n$. Then, since the nilradical is an ideal, $f - \sum_{i=0}^{n-1} a_i x^i = \sum_{i=n}^{\infty} a_i x^i$ is nilpotent. Thus, there is some m such that

$$0 = \left(\sum_{i=n}^{\infty} a_i x^i \right)^m \equiv a_n^m x^{nm} \mod (x^{nm+1}),$$

so a_n is nilpotent. The result follows by induction. □

Remark. *Consider the direct product ring $A = \Pi_{i=2}^{\infty} (\mathbb{Z}/2^i\mathbb{Z})$. Then, construct*

$$f = \sum_i a_i x^i,$$

where $a_i = (2, \ldots, 2, 0, \ldots)$; 2 appears until slot i, whereafter it is 0 indefinitely. Let $a_0 = a_1 = 0$ since there is no map into the ring. So, the minimal exponent that annihilates a_i is i whenever $i > 1$. Each a_i is therefore nilpotent.

Note that we have principal ideals $\mathfrak{q}_n = (1, \ldots, 1, 0, \ldots)$ that project the ring onto $\Pi_{i=2}^{n} (\mathbb{Z}/2^i\mathbb{Z})$; $a_i \in \mathfrak{q}_i$.

On the other hand, for any m, consider f^m. Then, consider $a_{m(m+1)}$. Note that the coefficient summands are the ordered partitions of $m(m+1)$ that have size m. There is precisely one summand that is not in the ideal $\mathfrak{q}_m$ and that is a_{m+1}^m, so the projection of the $x^{m(m+1)}$ coefficient onto $A/\mathfrak{q}_m$ is a_{m+1}^m. But, this is nonzero since a_{m+1}^m projects onto $\mathbb{Z}/2^{m+1}\mathbb{Z}$ as 2^m. Thus, f^m has a nonzero $x^{m(m+1)}$ coefficient.

In particular, for general commutative rings with unit (not necessarily Noetherian), if a_i is nilpotent for all i, this does not imply that $f = \sum_{i=0}^{\infty} a_i x^i$ is nilpotent.

(iii) Proof. Consider the ring homomorphism $\varphi : A[[x]] \to A$ that sends the power series to its constant term.

For every maximal ideal $\mathfrak{m}$ of A, $\varphi^{-1}(\mathfrak{m}) = \mathfrak{m} + xA[[x]]$ is an ideal, the contraction of $\mathfrak{m}$. Each of these are contained in maximal ideals of $A[[x]]$, so

$$\bigcap_{\mathfrak{m}} (\mathfrak{m} + xA[[x]]) \subset \mathfrak{R}(A[[x]])$$
$$\bigcap_{\mathfrak{m}} \mathfrak{m} + xA[[x]] \subset \mathfrak{R}(A[[x]])$$
$$\mathfrak{R}(A) + xA[[x]] \subset \mathfrak{R}(A[[x]])$$

In particular, if $a_0 \in \mathfrak{R}(A)$ and f has constant term a_0, then $f \in \mathfrak{R}(A[[x]])$.

Conversely, assume a_0 is not in the Jacobson radical of A. Then, there is some maximal ideal $\mathfrak{m}$ in A that does not contain a_0. But, $\mathfrak{m} + xA[[x]]$ is a non-trivial ideal of $A[[x]]$ under the contraction $\varphi^{-1}(\mathfrak{m})$. If $\mathfrak{m} + xA[[x]]$ is maximal, we'd have that f is not in the maximal ideal $\mathfrak{m} + xA[[x]]$ (since its constant term $a_0 \notin \mathfrak{m}$), so $f \notin \mathfrak{R}(A[[x]])$.

So, the proof is complete once we show that $\mathfrak{m} + xA[[x]]$ is maximal. Let $\mathfrak{r}$ be the maximal ideal containing $\mathfrak{m} + xA[[x]]$. In order for these ideals to be unequal, there is some $g \in \mathfrak{r}$ such that $g_0 \notin \mathfrak{m}$. However, projecting to A (inverse map of homomorphism $A \hookrightarrow A[[x]]$), we have an ideal $\mathfrak{r}'$ containing $\mathfrak{m}$ properly, so by the maximality of $\mathfrak{m}$, $\mathfrak{r} = A$. Thus, $\mathfrak{r} = A + xA[[x]] = A[[x]]$, violating maximality, so $\mathfrak{m} + A[[x]]$ is maximal. □

(iv) Proof. First, let's show that for any maximal ideal $\mathfrak{m}$ of A, $\mathfrak{m} + xA[[x]]$ is maximal. Let $\mathfrak{r}$ be the maximal ideal containing $\mathfrak{m} + xA[[x]]$. In order for these ideals to be unequal, there must be some $g \in \mathfrak{r}$ such that $g_0 \notin \mathfrak{m}$. However, projecting to A (inverse map of homomorphism $A \hookrightarrow A[[x]]$), we have an ideal $\mathfrak{r}'$ containing $\mathfrak{m}$ properly, so by the maximality of $\mathfrak{m}$, $\mathfrak{r} = A$. Thus, $\mathfrak{r} = A + xA[[x]] = A[[x]]$, violating maximality, so $\mathfrak{m} + A[[x]]$ is maximal.

Suppose $\mathfrak{m}$ is a maximal ideal of $A[[x]]$. Then, under the ring homomorphism $\varphi : A \hookrightarrow A[[x]]$, $\mathfrak{m}^c = \varphi^{-1}(\mathfrak{m})$ is an ideal of A. First of all, if $\mathfrak{m}^c = A$, then $\mathfrak{m}$ contains some series with constant term 1; by (i), it contains a unit, so $\mathfrak{m} = A[[x]]$, violating maximality.

If $\mathfrak{m}^c = 0$, then $\mathfrak{m}$ is contained in the ideal $xA[[x]]$, so $\mathfrak{m} = xA[[x]]$. Note that $xA[[x]]$ is maximal when A only has the ideals 0 and (1) (i.e., A is a field) since for any ideal I of A (not 0 or A), $I + xA[[x]]$ is a proper ideal containing $\mathfrak{m}$, violating the maximality of $\mathfrak{m}$. Otherwise, the only ideal containing $xA[[x]]$ is $A + xA[[x]] = A[[x]]$, the whole ring. For our purposes, just note that $\mathfrak{m}$ is indeed generated by x and $\mathfrak{m}^c$.

Thus, we are left with the case where $\mathfrak{m}^c$ is neither 0 nor A. Since it is a nontrivial ideal, it is contained in some maximal ideal $\mathfrak{r}$ of A. Thus, $\mathfrak{m} \subset \mathfrak{m}^c + xA[[x]] \subset \mathfrak{r} + xA[[x]]$. By the maximality of $\mathfrak{m}$, $\mathfrak{m}^c = \mathfrak{r}$, and $\mathfrak{m} = \mathfrak{m}^c + xA[[x]]$, completing the proof. □

(v) Proof. Let $\mathfrak{p}$ be a prime ideal of A. Consider the ring homomorphism $\varphi : A[[x]] \to A$ that maps each series to its constant term. Then, $\varphi^{-1}(\mathfrak{p}) = \mathfrak{p} + xA[[x]]$ is a prime ideal of $A[[x]]$ since contraction preserves primality. Under the ring homomorphism $\psi : A \hookrightarrow A[[x]]$, $\mathfrak{p}$ is the contraction of the prime ideal $\mathfrak{p} + xA[[x]]$, proving the claim. □

1.6 Exercise

Proof. Since all maximal ideals are prime, we know that $\mathfrak{N} \subset \mathfrak{R}$. Suppose to the contrary that the ideal $\mathfrak{R}$ is not contained in $\mathfrak{N}$. Then, by assumption, there is some nonzero idempotent $e \in \mathfrak{R}$.

Since $e^2 = e$, a simple inductive argument reveals that e cannot be nilpotent.

Let $\mathfrak{p}$ be any prime ideal. Consider the map $A \to A/\mathfrak{p}$. Because $A/\mathfrak{p}$ is an integral domain and $0 = \overline{e^2 - e} = \overline{e}(\overline{e} - 1)$, we have $e \in \mathfrak{p}$ or $e \in 1 + \mathfrak{p}$. If $e \in \mathfrak{p}$ for all prime ideals, then e must also be in nilradical, as it is the intersection of prime ideals. Thus, there is some prime ideal $\mathfrak{p}'$ such that $e \in 1 + \mathfrak{p}'$.

There is some maximal ideal $\mathfrak{m}'$ containing $\mathfrak{p}'$, so $e \in 1 + \mathfrak{m}'$. However, since $e \in \mathfrak{R}$, which is the intersection of all maximal ideals, $e \in \mathfrak{m}'$ also. Together, these two facts imply $1 \in \mathfrak{m}'$, yielding the desired contradiction. □

1.7 Exercise

Proof. Let $\mathfrak{p}$ be a prime ideal so that $A/\mathfrak{p}$ is an integral domain. For any $y \in A/\mathfrak{p}$ such that $y \neq 0$, there is some $x \in A$, such that $\overline{x} = y$. By assumption, there is an $n > 1$ such that $x^n = x$, so $y^n = y$. Note that $y^{n-1} - 1 \in A/\mathfrak{p}$ and

$$y(y^{n-1} - 1) = y^n - y = 0.$$

Since $A/\mathfrak{p}$ is an integral domain and $y \neq 0$, $y^{n-1} = 1$. In particular, y is a unit. Thus, all nonzero elements are nonzero, implying that $A/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. □

1.8 Exercise

Proof. Let $\mathcal{S}$ be the set of prime ideals of $\mathcal{S}$. If $\mathfrak{p} \subset \mathfrak{q}$, we'll define a partial ordering by $\mathfrak{q} \leq \mathfrak{p}$. Now, suppose we have a chain $\mathcal{T} \subset \mathcal{S}$. Let

$$Q = \bigcap_{\mathfrak{q} \in \mathcal{T}} \mathfrak{q},$$

which, as an intersection of ideals, is itself an ideal.

Suppose $a, b \in A$ and $ab \in Q$. If $a \notin Q$, then there is some $\mathfrak{r} \in \mathcal{T}$ such that $a \notin \mathfrak{r}$. Since $a \notin \mathfrak{r}$, for any $\mathfrak{q}$ such that $\mathfrak{q} \subset \mathfrak{r}$, $a \notin \mathfrak{q}$. Since $\mathcal{T}$ is a chain,

$$Q = \bigcap_{\mathfrak{q} \leq \mathfrak{r}} \mathfrak{q} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \mathfrak{r} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q}.$$

Let $\mathfrak{q} \geq \mathfrak{r}$. Then, since $ab \in Q \subset \mathfrak{q}$ and $\mathfrak{q}$ is prime, we must have $b \in \mathfrak{q}$, as we've already shown a cannot be in $\mathfrak{r}$. Thus, $b \in \mathfrak{q}$ for all $\mathfrak{q} \geq \mathfrak{r}$, so $b \in Q$. Thus, Q is a prime ideal.

For all $\mathfrak{q} \in \mathcal{T}$, $Q \subset \mathfrak{q}$ ($Q \geq \mathfrak{q}$) and is thus a maximal element. Using Zorn's lemma, $\mathcal{S}$ has at least one maximal element. By our notation, these are minimal elements with respect to inclusion since we used $\geq$ for $\subset$. □

1.9 Exercise

Proof. If $\mathfrak{a} = \bigcap_{\mathfrak{p}} \mathfrak{p}$ is an intersection of prime ideals, then

$$r(\mathfrak{a}) = r\left(\bigcap_{\mathfrak{p}} \mathfrak{p}\right) = \bigcap_{\mathfrak{p}} r\left(\mathfrak{p}\right) = \bigcap_{\mathfrak{p}} \mathfrak{p} = \mathfrak{a}.$$

Conversely, suppose $r(\mathfrak{a}) = \mathfrak{a}$. By Proposition 1.14, $r(\mathfrak{a}) = \bigcap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over the prime ideals containing $\mathfrak{a}$. Therefore, $\mathfrak{a}$ is an intersection of prime ideals. □

1.10 Exercise

Proof. (i) $\implies$ (ii). Since $\mathfrak{N}$ is the intersection of all prime ideals, $\mathfrak{N}$ is the unique prime ideal. By Theorem 1.3, A must have a maximal ideal, but since every maximal ideal is prime and $\mathfrak{N}$ is the only prime ideal, $\mathfrak{N}$ must be maximal. Hence, it is also the only maximal ideal. By Corollary 1.5, every non-unit is contained in a maximal ideal, so every non-unit must be in $\mathfrak{N}$, hence nilpotent.

(ii) $\implies$ (iii). Consider the ring homomorphism $A \to A/\mathfrak{N}$. Every element $x \in A$ is either a unit or nilpotent; if x is a unit, $\overline{x}$ is a unit, and if x is nilpotent, $x \in \mathfrak{N}$, so $\overline{x} = 0$. Since the projection is surjective, every nonzero element of the ring $A/\mathfrak{N}$ is a unit, i.e. $A/\mathfrak{N}$ is a field.

(iii) $\implies$ (i). Since $A/\mathfrak{N}$ is a field, $\mathfrak{N}$ a maximal ideal. All maximal ideals are prime. However, $\mathfrak{N}$ is also the intersection of all prime ideals. Suppose there is another prime ideal $\mathfrak{p}$, not equal to $\mathfrak{N}$. Thus, there is an element $x \in \mathfrak{p} - \mathfrak{N}$ since $\mathfrak{N} \subset \mathfrak{p}$. Under the map $A \to A/\mathfrak{N}$, $\mathfrak{x}$ is a unit since it is non-zero. There is a y in A such that $xy = 1 + \mathfrak{N}$. But, $xy \in \mathfrak{p}$ since $x \in \mathfrak{p}$, so $\mathfrak{p} + \mathfrak{N} = (1)$ and are thus coprime. But $\mathfrak{N} \subset \mathfrak{p}$, so $\mathfrak{p} = (1)$, a contradiction. □

1.11 Exercise

(i) Proof. For every $x \in A$,

$$\begin{aligned}(1+x)^2 &= 1+x\\ 1+2x+x^2 &= 1+x\\ 1+2x+x &= 1+x\\ 2x &= 0.\end{aligned}$$

□

(ii) Proof. Suppose $x \in A$ represents the class $\bar{x} \in A/\mathfrak{p}$. Then,

$$\bar{x}^2 = (\mathfrak{p}+x)(\mathfrak{p}+x) = \mathfrak{p}+x^2 = \mathfrak{p}+x = \bar{x}.$$

Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain. Hence, $\bar{x}^2 = \bar{x} \implies \bar{x}(\bar{x}-1) = 0 \implies \bar{x} \in \{0,1\}$. Thus, $A/\mathfrak{p}$ is the field of size 2, making $\mathfrak{p}$ a maximal ideal. □

(iii) Proof. For I finitely-generated, $I = (x_1, \ldots, x_n)$ for $x_i \in A$. We'll show that I is principal.

We do this by induction, proving that an ideal with n-many generators can be generated by a single element. The base case is trivial. For the inductive step, suppose that for all ideals generated by $n-1$ elements (or fewer), $(a_1, \ldots, a_{n-1}) = (a)$. Consider $(x_1, \ldots, x_n)$. By the inductive step, $(x_1, \ldots, x_{n-1}) = (a)$, so

$$(x_1, \ldots, x_n) = (a, x_n).$$

Consider the element $b = a + x_n + ax_n$. Note that

$$ab = a^2 + ax_n + a^2x_n = a + ax_n + ax_n = a + 2ax_n = a$$

and

$$x_nb = ax_n + x_n^2 + ax_n^2 = ax_n + x_n + ax_n = x_n + 2ax_n = x_n.$$

Thus, $a, x_n \in (b)$, so $(x_1, \ldots, x_n) = (a, x_n) = (b)$ since, by the construction of b, $b \in (a, x_n)$. By induction, any finitely generated ideal of a Boolean ring A is principal. □

1.12 Exercise

Proof. If x is a unit, $x^2 = x \implies x^2x^{-1} = xx^{-1} \implies x = 1$. If x is not a unit, then $x \in \mathfrak{m}$. That means that $1 - x$ is not in $\mathfrak{m}$ (otherwise $x + (1 - x) = 1 \in \mathfrak{m}$) and consequently that $1 - x$ is a unit. $x^2 = x \implies x(1 - x) = 0$, but since $1 - x$ is a unit, we have $x = 0$. □

1.13 Exercise

Here, we construct an algebraic closure of a field K.

Let K be a field, and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. We now show that $\mathfrak{a} \neq (1)$.

Proof. Let R be any nonzero commutative ring with unit. Consider $R[x]/(f(x))$, where $f(x)$ is a monic polynomial over R. Suppose $R[x] = (f(x))$. Then, $1 = f(x)r(x)$ for some $r(x) \in R[x]$. However, by Exercise 1.2(i), $f(x)$ cannot be a unit since its highest coefficient is 1, a unit. In particular, $(f(x))$ is a proper ideal of $R[x]$. Thus, for a nonzero ring R, $R[x]/(f(x))$ is also nonzero.

We'll show that $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ is nonzero if $\{f_i\}$ are monic polynomials over K. We do this by induction. We proved the base case above, setting $K = R$.

Now, suppose $K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1}))$ is nonzero. Then,

$$\begin{aligned} K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n)) &\cong K[x_1, \ldots, x_{n-1}][x_n]/(f_1(x_1), \ldots, f_n(x_n)) \\ &\cong (K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1})))\,[x_n]/(f_n(x_n)) \\ &\cong R[x_n]/(f_n(x_n)), \end{aligned}$$

where $R = K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1}))$. Since R is nonzero and $f_n(x_n)$ is a monic polynomial over R, we have that $R[x_n]/(f_n(x_n))$ is nonzero by our initial proof. By induction, $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ is nonzero.

We turn to the main point. Suppose to the contrary that $\mathfrak{a} = (1)$. Then, there is some finite collection of pairs $(h_i, f_i) \in A \times \Sigma$ such that

$$1 = \sum_{i=1}^{n} h_i f_i(x_{f_i}).$$

Write $x_i = x_{f_i}$ for convenience. For the indeterminates x_g not equal to some f_i, we can map $x_g \mapsto 0$. This is a ring homomorphism of A. This maps the $h_i \mapsto k_i \in K[x_1, \ldots, x_n]$ so that

$$1 = \sum_{i=1}^{n} k_i f_i(x_i).$$

But then, we have $(f_1(x_1), \ldots, f_n(x_n)) = (1)$, violating $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ being nonzero. Thus, $\mathfrak{a} \neq (1)$. □

By Corollary 1.4, A has a maximal ideal $\mathfrak{m}$ containing $\mathfrak{a}$ since $\mathfrak{a} \neq (1)$. Now, let $K_1 = A/\mathfrak{m}$, which is a field since $\mathfrak{m}$ is maximal. Since $\mathfrak{m}$ cannot contain any units and $K \hookrightarrow A$, K is a subfield of K_1. Further, for every $f \in \Sigma$, $x_f \in K_1$ and satisifies $f(x_f) = 0$.

Since K_1 is a field, we can repeat the construction with K_1 in place of K, obtaining a field K_2, and so on. Now, let $L = \bigcup_{n=1}^{\infty} K_n$, which is a field since all operations take place in some subfield K_n.

Suppose $f \in \Sigma$ has degree n. In K_1, if x_f maps to the residue class α_f in $A/\mathfrak{m}$, then $f(x) = (x - \alpha_f)g(x)$, where $g(x) \in K_1[x]$ has degree $n-1$. In the worse case, g is irreducible here. By a simple inductive argument, $f(x)$ factors completely over $K_n \subset K$. Thus, every $f \in \Sigma$ splits completely in K.

By the same logic, let $f \in L[x]$ be a monic polynomial; suppose f has degree n. Since f has finitely-many coefficients, all coefficients lie in some common subfield K_k. Then, by construction, f splits completely in K_{k+n}, so L is algebraically closed.

Now, let $\overline{K}$ be the set of all elements of L that are algebraic over K. Then, $\overline{K}$ is an algebraic closure of K.

1.14 Exercise

Proof. Since 0 is such an ideal, Σ is non-empty. For $I, J \in \Sigma$, we say that $I \leq J$ if $I \subset J$, endowing Σ with a partial ordering. Let $\mathcal{T}$ be a chain of Σ. Then, for all $I \in \mathcal{T}$, let $T = \bigcup_{I \in \mathcal{T}} I$. T is an ideal (because the chain is ordered by inclusion) and contains only zero-divisors, so $T \in \Sigma$. Further, $I \subset T$ for all $I \in \mathcal{T}$. Thus, every chain has an upper bound in Σ. By Zorn's lemma, Σ has at least one maximal element.

Now that we've proven existence, let M be a maximal element of Σ. We now show that M is prime. Note that the elements of A that are not zero divisors form a multiplicatively-closed set S. Suppose $x, y \in A - M$. By the maximality of M, $M + (x)$ and $M + (y)$ must both contain elements of S. Thus, there are $m_1, m_2 \in M$ and $a_1, a_2 \in A$ so that $m_1 + a_1 x, m_2 + a_2 y \in S$. Since S is multiplicatively-closed,

$$(m_1 + a_1 x)(m_2 + a_2 y) = m_1 m_2 + m_1 a_2 y + m_2 a_1 x + a_1 a_2 xy \in S.$$

If $xy \in M$, then all terms would be in M so that the sum would be in M. However, M only contains zero divisors and S contains no zero divisors, so this would be a contradiction. Thus, $xy \notin M$, proving the primality of M.

If z is a zero divisor, then $(z) \in \Sigma$. Thus, (z) is contained in some maximal element M of Σ, which we've just shown is prime. Thus, the set of zero divisors in A is a union of prime ideals. □

1.15 Exercise

(i) Proof. Clearly, $V(\mathfrak{a}) \subset V(E)$ since $E \subset \mathfrak{a}$. However, if $\mathfrak{p}$ contains E, then the ideal $\mathfrak{a}$ that E generates must also be contained in $\mathfrak{p}$, so $V(E) = V(\mathfrak{a})$.

Since $\mathfrak{a} \subset r(\mathfrak{a})$, we have $V(r(\mathfrak{a})) \subset V(\mathfrak{a})$. Let $\mathfrak{p}$ be any prime containing $\mathfrak{a}$; consider the ring homomorphism $\varphi : A \to A/\mathfrak{p}$. Then, $\varphi(\mathfrak{a}) = 0$. If $x \in r(\mathfrak{a})$, then $x^n \in \mathfrak{a}$ for some $n > 0$. Under the map, we have $\overline{x}^n = 0$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain, so $\overline{x} = 0$. Thus, $x \in \mathfrak{p}$ and $r(\mathfrak{a}) \subset \mathfrak{p}$. Consequently, $V(\mathfrak{a}) = V(r(\mathfrak{a}))$. □

(ii) Proof. $V(0) = X$ since for any prime ideal $\mathfrak{p} \in X$, $0 \subset \mathfrak{p}$. Likewise, for any prime ideal $\mathfrak{p} \in X$, $\mathfrak{p}$ is a proper ideal of A, so $\mathfrak{p}$ does not contain (1). Thus, $V(1) = \emptyset$. □

(iii) Proof. Let $(E_i)_{i\in I}$ be a family of subsets of A. Suppose $\mathfrak{p}$ is a prime ideal containing $\bigcup_{i\in I} E_i$. Then, $\mathfrak{p}$ contains each E_i; i.e.,

$$V\left(\bigcup_{i\in I} E_i\right) \subset \bigcap_{i\in I} V(E_i).$$

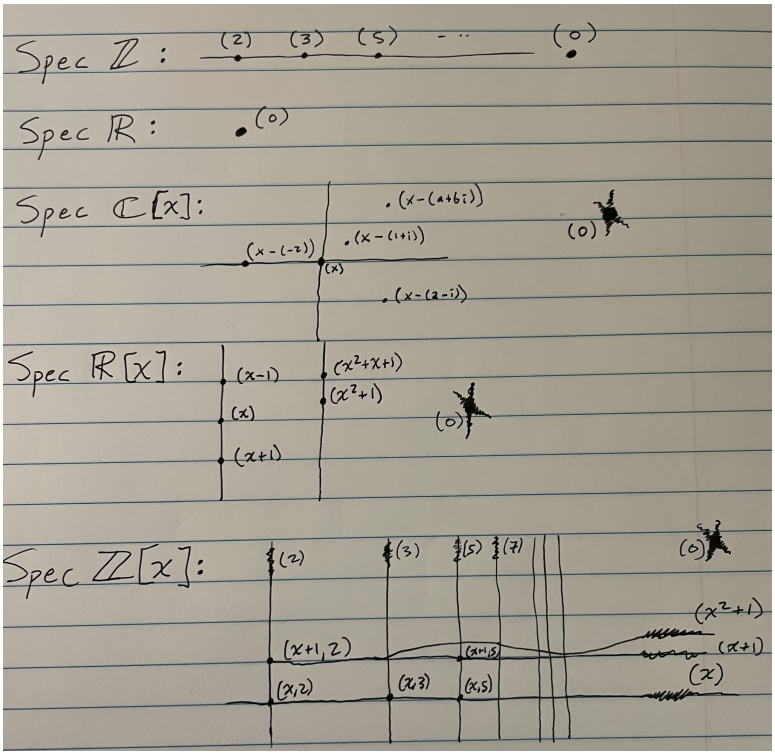
Conversely, suppose $\mathfrak{p}$ contains each E_i; then clearly, it contains the union. So, we have equality. □

(iv) Proof. If a prime ideal $\mathfrak{p} \in V(\mathfrak{a})\cup V(\mathfrak{b})$, then it is in one of the two sets. Assume without loss of generality that $\mathfrak{p} \in V(\mathfrak{a})$. Then, $\mathfrak{p}$ contains $\mathfrak{a}$ and thus also contains $\mathfrak{a}\cap\mathfrak{b}$, so $V(\mathfrak{a})\cup V(\mathfrak{b}) \subset V(\mathfrak{a}\cap\mathfrak{b})$. Since $\mathfrak{a}\mathfrak{b} \subset \mathfrak{a}\cap\mathfrak{b}$, we have $V(\mathfrak{a}\cap\mathfrak{b}) \subset V(\mathfrak{a}\mathfrak{b})$. Thus, if we can show that $V(\mathfrak{a}\mathfrak{b}) \subset V(\mathfrak{a})\cup V(\mathfrak{b})$, we will have shown full equality.

Let $\mathfrak{p}$ be a prime ideal containing $\mathfrak{a}\mathfrak{b}$. Consider the homomorphism $\varphi : A \to A/\mathfrak{p}$. Since φ is surjective, the ideals $\mathfrak{a}$ and $\mathfrak{b}$ map to ideals $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{b}}$ of $A/\mathfrak{p}$. Suppose to the contrary that both $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{b}}$ contain nonzero elements, say a and b. Then, ab is nonzero since $A/\mathfrak{p}$ is an integral domain ($\mathfrak{p}$ is prime). However, $\overline{\mathfrak{a}}\cdot\overline{\mathfrak{b}} = \overline{\mathfrak{a}\mathfrak{b}} = 0$ since $\mathfrak{a}\mathfrak{b} \subset \mathfrak{p}$. Thus, either $\overline{\mathfrak{a}} = 0$ or $\overline{\mathfrak{b}} = 0$. In other words, $\mathfrak{a} \subset \mathfrak{p}$ or $\mathfrak{b} \subset \mathfrak{p}$. In either case, $\mathfrak{p} \in V(\mathfrak{a}) \cup V(\mathfrak{b})$, completing the proof. □

Remark. *We've shown that $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the Zariski topology. We denote this topological space* $\mathrm{Spec}(A) := X$*, the prime spectrum of A.*

1.16 Exercise



1.17 Exercise

Before proving properties (i)-(vii), we must show that the sets X_f form a basis of open sets for the Zariski topology.

Proof. Any open set U, by definition, is the complement of a closed set V of $\operatorname{Spec} A$. In Exercise 1.15, we showed that each closed set of $\operatorname{Spec} A$ can be represented by $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ (since arbitrary intersections and finite unions have the same form), so $U = V(\mathfrak{a})^c$ for some ideal $\mathfrak{a}$. By Exercise 1.15(iii),

$$V(\mathfrak{a}) = V\left(\bigcup_{f\in\mathfrak{a}}\right) = \bigcap_{f\in\mathfrak{a}} V(f).$$

Thus,

$$U = V(\mathfrak{a})^c = \left(\bigcap_{f\in\mathfrak{a}} V(f)\right)^c = \bigcup_{f\in\mathfrak{a}} V(f)^c = \bigcup_{f\in\mathfrak{a}} X_f,$$

so $\{X_f\}_{f\in A}$ forms a basis of open sets in $\operatorname{Spec} A$. □

(i) Proof. Note that $V(f) = V(\{f\})$, the prime ideals containing the element $f \in A$. Then, $X_f = V(\{f\})^c$, where $-^c$ denotes the complement. By Exercise 1.15(i), $V(f) = V((f))$ for any $f \in A$. If $f, g \in A$, then using Exercise 1.15(iv), we have

$$X_f \cap X_g = V((f))^c \cap V((g))^c = (V((f)) \cup V((g)))^c = V((f)(g))^c = V((fg))^c = V(fg)^c = X_{fg},$$

completing the proof. □

(ii) Proof. If $X_f = \emptyset$, then $V(f) = X$. Thus, f is in the intersection of all prime ideals, meaning $f \in \mathfrak{N}_A$ (i.e., f is nilpotent). Conversely, if f is nilpotent, it is in the nilradical and thus in every prime ideals, so $X_f = V(f)^c = X^c = \emptyset$. □

(iii) Proof. If $X_f = X$, then $V(f) = \emptyset$. Thus, f is not in any prime ideal and thus not in any maximal ideal. In particular, $(f) = A$, so f is a unit. Conversely, if f is a unit, then there is some $g \in A$ such that $fg = 1$. Thus, for any prime ideal $\mathfrak{p}$, under $A \to A/\mathfrak{p}$, we have $\overline{f} \cdot \overline{g} = 1$. So, $\overline{f} \neq 0$ and $f \notin \mathfrak{p}$. Hence, $V(f) = \emptyset$ and $X_f = X$. □

(iv) Proof. If $X_f = X_g$, then $V(f) = V(g)$. By Proposition 1.14,

$$r(f) = \bigcap_{\mathfrak{p}:(f)\subset\mathfrak{p}} \mathfrak{p} = \bigcap_{\mathfrak{p}\in V(f)} \mathfrak{p} = \bigcap_{\mathfrak{p}\in V(g)} \mathfrak{p} = r(g).$$

Conversely, if $r(f) = r(g)$, then $V(f) = V(r(f)) = V(r(g)) = V(g)$, so $X_f = X_g$. □

(v) Proof. Suppose $\{U_j\}_{j\in J}$ is an open cover of X. Since we showed that $\{X_f\}_{f\in A}$ form a basis of open sets, we can write $U_j = \bigcup_{f\in I_j} X_f$ for each $j \in J$. Thus, if $I = \bigcup_{j\in J} I_j$, we have an open cover

$$X = \bigcup_{f\in I} X_f = \bigcup_{f\in I} V(f)^c = \left(\bigcap_{f\in I} V(f)\right)^c$$

$$\implies \emptyset = \bigcap_{f\in I} V(f) = V\left(\bigcup_{f\in I}(f)\right),$$

using Exercise 1.15(iii). Let $\mathfrak{a}$ be the ideal generated by $f \in I$. Then, $V(\mathfrak{a}) = \emptyset$, so $V(r(\mathfrak{a})) = \emptyset$ by Exercise 1.15(i). No prime ideal contains $r(\mathfrak{a})$, so $r(\mathfrak{a}) = (1)$. Thus, $\mathfrak{a} = (1)$.

In particular, there is some finite collection $\{f_s\}_{s\in S}$ of $\{f_i\}_{i\in I}$ such that

$$\sum_{s\in S} a_s f_s = 1$$

for some $a_s \in A$. Thus, there is no prime ideal that divides every (f_i), so $\bigcap_{s\in S} V(f_s) = \emptyset$ and $X = \bigcup_{s\in S} X_{f_s}$. Since each f_s is contained in some U_s, we have

$$X = \bigcup_{s\in S} U_s,$$

where S is finite and $U_s \in \{U_j\}_{j\in J}$. Hence, X is quasi-compact. □

(vi) Proof. For $f \in A$, let $\{U_j\}_{j\in J}$ be an open covering of X_f. Then, for each j, $U_j = \bigcup_{g\in I_j} X_g$ since $\{X_g\}_{g\in A}$ form a basis of open sets. If $I = \bigcup_{j\in J} I_j$, then

$$X_f \subset \bigcup_{g\in I} X_g.$$

Taking the intersection with X_f and using (i), we have

$$X_f \subset X_f \cap \bigcup_{g\in I} X_g = \bigcup_{g\in I} (X_f \cap X_g) = \bigcup_{g\in I} X_{fg} \subset X_f,$$

so $X_f = \bigcup_{g\in I} X_{fg}$. Taking complements and using results of Exercise 1.15, we have

$$V(f) = \bigcap_{g\in I} V(fg) = V\left(\bigcup_{g\in I} (fg)\right) = V\left(\sum_{g\in I} (fg)\right).$$

By (iv), $r(f) = r\left(\sum_{g\in I} fg\right)$. Therefore, for some power $n > 0$ of f, we have

$$f^n = \sum_{s\in S} a_s f g_s$$

for a finite subset $S \subset I$ and $a_s \in A$. From the sum, if there is some prime ideal $\mathfrak{p}$ that does not contain f (and thus not f^n), it does not contain at least one element $a_s f g_s$ on the right-hand side. So, it does not contain g_s for some $s \in S$. In particular,

$$V(f)^c \subset \bigcup_{s\in S} V(g_s)^c$$
$$X_f \subset \bigcup_{s\in S} X_{g_s}.$$

Since each X_{g_s} is contained in some $U \in \{U_j\}_{j\in J}$ by the definition of I, X_f has a finite subcover and is thus quasi-compact. □

(vii) Proof. Suppose U is an open set of X and that U is a finite union of sets X_f. Now, let S be an open covering of U. Then, it is an open covering of each X_f. By (vi), each X_f has a finite subcover. The union of these finite subcovers is finite and covers U, so U is quasi-compact.

Conversely, suppose U is quasi-compact. By Exercise 1.15, $U = V(\mathfrak{a})^c$ for some ideal $\mathfrak{a}$. Thus, by Exercise 15(iii),

$$U = V(\mathfrak{a})^c = V\left(\bigcup_{f\in\mathfrak{a}} (f)\right)^c = \left(\bigcap_{f\in\mathfrak{a}} V(f)\right)^c = \bigcup_{f\in\mathfrak{a}} V(f)^c = \bigcup_{f\in\mathfrak{a}} X_f.$$

Since U is quasi-compact, there is a finite subcover, so U is the union of finitely-many X_f. □

1.18 Exercise

(i) Proof. Let $x \in \operatorname{Spec} A$. If $\mathfrak{p}_x$ is maximal, then $V(\mathfrak{p}_x)$ contains only the prime $\mathfrak{p}_x$. Otherwise, there is a prime ideal containing $\mathfrak{p}_x$ properly, violating maximality. So in this case, x is a closed point ($\{x\}$ is closed). Conversely, if $\mathfrak{p}_x$ is not maximal, there is some maximal ideal $\mathfrak{p}_y$ properly containing $\mathfrak{p}_x$. In order for $\{x\}$ to be closed, there must be some subset $E \subset A$ such that $V(E) = \{x\}$. But in this case, $E \subset \mathfrak{p}_x \subset \mathfrak{p}_y$, so $\{x, y\} \subset V(E)$. Thus, x is not a closed point when $\mathfrak{p}_x$ is not maximal. □

(ii) Proof. Any closed set containing x is, by definition, of the form $V(E)$ for some subset $E \subset A$, and by assumption, $x \in V(E)$. Thus, $E \subset \mathfrak{p}_x$. By Exercise 1.15(iii),

$$\overline{\{x\}} = \bigcap_{E \subset \mathfrak{p}_x} V(E) = V\left(\bigcup_{E \subset \mathfrak{p}_x} E\right) = V(\mathfrak{p}_x).$$

□

(iii) Proof. By (ii), $\overline{\{x\}} = V(\mathfrak{p}_x)$, so we need to show that $y \in V(\mathfrak{p}_x) \iff \mathfrak{p}_x \subset \mathfrak{p}_y$. But, this holds by definition. □

(iv) Proof. We aim to show that $\operatorname{Spec} A$ is T_0, so for any two elements $x, y \in \operatorname{Spec} A$, we must show that there is an open set that contains only one of x and y. Equivalently, we can produce a closed set that contains exactly one.

Let $x, y \in \operatorname{Spec} A$ be distinct points. If $y \notin V(\mathfrak{p}_x)$, then we are done since $x \in V(\mathfrak{p}_x)$. So, suppose $y \in V(\mathfrak{p}_x)$. By (iii), $\mathfrak{p}_x \subset \mathfrak{p}_y$. Now, consider $V(\mathfrak{p}_y)$. If $x \notin V(\mathfrak{p}_y)$, then we are done since $y \in V(\mathfrak{p}_y)$. So, suppose $x \in V(\mathfrak{p}_y)$. By (iii), $\mathfrak{p}_y \subset \mathfrak{p}_x$, so $\mathfrak{p}_x = \mathfrak{p}_y$ (or $x = y$), providing the necessary contradiction. □

1.19 Exercise

Proof. A topological space is irreducible if $X \neq \emptyset$ and every pair of non-empty open sets in X intersect. Since $\{X_f\}_{f \in A}$ form an open basis of X, X is irreducible if and only if $X_f \cap X_g \neq \emptyset$ for all $f, g \in A$ satisyfing $X_f \neq \emptyset$ and $X_g \neq \emptyset$.

Suppose that the nilradical $\mathfrak{N}$ is prime, and suppose $f, g \in A$ satisfy $X_f \neq \emptyset$ and $X_g \neq \emptyset$. Then, $V(f) \neq X$ and $V(g) \neq X$. Since $\mathfrak{N}$ is the intersect of all primes in X, $\mathfrak{N} \notin V(f) \cup V(g)$ since, by assumption, neither $V(f)$ nor $V(g)$ contain all primes. Thus, $\mathfrak{N} \in V(f)^c \cap V(g)^c = X_f \cap X_g$, so $X_f \cap X_g$ is non-empty.

If the nilradical is not prime, then there are $f, g \in A$ such that $fg \in \mathfrak{N}$ yet neither f nor g are nilpotent. By Exercise 1.17(ii), X_f and X_g are thus non-empty. However, by Exercise 1.17(i), $X_f \cap X_g = X_{fg}$. Since $fg \in \mathfrak{N}$ and $\mathfrak{N}$ is the intersection of all prime ideals, each prime ideal contains fg, so $V(fg) = X$ and $X_{fg} = \emptyset$. $X_f \cap X_g = \emptyset$ even though both X_f and X_g are non-empty, so X is not irreducible. □

1.20 Exercise

(i) Proof. Let Y be an irreducible subspace of X. Because $\overline{Y}$ is closed, any closed set V of X yields a closed set $V \cap \overline{Y}$ in X. Since all closed sets of the subspace $\overline{Y}$ have this form (definition of subspace topology), any closed set of $\overline{Y}$ is inherited from a closed set of X.

Since Y is irreducible, $Y \neq \emptyset$ (thus $\overline{Y} \neq \emptyset$) and every pair of non-empty open sets in Y intersect. Equivalently, every pair of closed sets V_1, V_2 not equal to all of Y satisfy $V_1 \cup V_2 \neq Y$.

Let V_1, V_2 be any two closed sets of $\overline{Y}$, neither alone containing all of $\overline{Y}$. Then, for some closed sets A_1, A_2 of X, we have $V_1 = A_1 \cap \overline{Y}$ and $V_2 = A_2 \cap \overline{Y}$. Thus, $B_1 = A_1 \cap Y$ and $B_2 = A_2 \cap Y$ are closed in Y. If, for instance, $A_1 \cap Y$ contains all of Y, then $A_1 \cap \overline{Y}$ is a closed set containing Y, so $\overline{Y} \subset A_1 \cap \overline{Y}$. But this is a contradiction since we assumed $A_1 \cap \overline{Y}$ does not contain all of $\overline{Y}$. Thus, neither B_1 nor B_2 alone contain all of Y. By the irreducibility of Y, $B_1 \cup B_2 \neq Y$.

Now, assume for the sake of contradiction that $V_1 \cup V_2 = \overline{Y}$. Then,

$$\overline{Y} = V_1 \cup V_2 = (A_1 \cap \overline{Y}) \cup (A_2 \cap \overline{Y}) = (A_1 \cup A_2) \cap \overline{Y}.$$

Since $Y \subset \overline{Y}$, we can intersect with Y:

$$Y = (A_1 \cup A_2) \cap \overline{Y} \cap Y = (A_1 \cup A_2) \cap Y = (A_1 \cap Y) \cup (A_2 \cap Y) = B_1 \cup B_2,$$

yielding our contradiction. Thus, $\overline{Y}$ is irreducible when Y is irreducible. □

(ii) Proof. Let Σ be the set of irreducible subspaces of X; $\subset$ gives Σ a partial ordering. Let $\mathcal{T}$ be a chain, and define $U = \bigcup_{Y \in \mathcal{T}} Y$. Let A_1, A_2 be two non-empty open sets of U. Choose some $x \in A_1$ and $y \in A_2$. Then, there is some Y_1 that contains x and some Y_2 that contains y. Since $\mathcal{T}$ is a chain, either $Y_1 \subset Y_2$ or $Y_2 \subset Y_1$. Without loss of generality, assume $Y_1 \subset Y_2$. Then, $x \in Y_1 \subset Y_2$, so $x, y \in Y_2$. For convenience, write $Y = Y_2$. Note that since A_1, A_2 are open in U, $A_1 = V_1 \cap U$ and $A_2 = V_2 \cap U$ for open sets V_1, V_2 in X. But also, $A_1^Y = V_1 \cap Y \subset A_1$ and $A_2^Y = V_2 \cap Y \subset A_2$ since $Y \subset U$. Since x is in A_1 and Y, $x \in A_1^Y$, and since y is in A_2 and Y, $y \in A_2^Y$, so both sets are open and non-empty. By the irreducibility of Y, $A_1^Y \cap A_2^Y \neq \emptyset$. Since $A_1^Y \subset A_1$ and $A_2^Y \subset A_2$, $A_1^Y \cap A_2^Y \subset A_1 \cap A_2$, so $A_1 \cap A_2$ is also non-empty. Thus, U is irreducible and $U \in \Sigma$. In particular, all chains in Σ have maximal elements in Σ. By Zorn's lemma, every irreducible subspace of X is contained in a maximal irreducible subspace. □

(iii) Proof. Let Y be a maximal irreducible subspace of X. By (i), $\overline{Y}$ is irreducible, but since Y is maximal and $Y \subset \overline{Y}$, we have $Y = \overline{Y}$, so Y is closed.

Let $x \in X$; consider the subspace $\{x\}$. If U_1 and U_2 are any non-empty open sets of $\{x\}$, then $U_1 = U_2 = \{x\}$. Thus, $U_1 \cap U_2 = \{x\} \neq \emptyset$. In particular, $\{x\}$ is an irreducible subspace. By (ii), it is contained in some maximal irreducible subspace. Thus, the maximal irreducible subspaces cover X.

Now, suppose X is Hausdorff. Let Y be an irreducible subspace of X, and let $x, y \in Y$ (Y is non-empty by irreducibility). If x and y are distinct, then since X is Hausdorff, there are open sets U_x and U_y such that $x \in U_x$, $y \notin U_x$, $y \in U_y$, $x \notin U_y$, and $U_x \cap U_y = \emptyset$. Thus, $Y \cap U_x$ and $Y \cap U_y$ are non-empty open sets of Y since they contain x and y, respectively. However,

$$(Y \cap U_x) \cap (Y \cap U_y) = Y \cap (U_x \cap U_y) = Y \cap \emptyset = \emptyset.$$

Thus, Y is not irreducible, yielding a contradiction. Thus, Y cannot have more than one point. Conversely, we've already shown that one-point sets are irreducible subspaces. Thus, the irreducible components of Hausdorff spaces are one-point sets. □

(iv) Proof. First, we show that $V(\mathfrak{p})$ is irreducible for any $\mathfrak{p} \in X$. In (iii), we showed that one-point sets are irreducible, i.e. $\{\mathfrak{p}\}$ is irreducible for every $\mathfrak{p} \in \operatorname{Spec} A$. Since $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ by Exercise 1.18(ii), (i) tells us that $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ inherits the irreducibility of $\{\mathfrak{p}\}$.

By (iii), every irreducible component is closed, so each is of the form $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$. If $\mathfrak{a} = (1)$, then $V(\mathfrak{a}) = \emptyset$, violating irreducibility. Thus, there is some prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subset \mathfrak{a}$, so $V(\mathfrak{a}) \subset V(\mathfrak{p})$. Since $V(\mathfrak{p})$ is irreducible, $V(\mathfrak{a}) = V(\mathfrak{p})$ by the maximality of $V(\mathfrak{a})$. Thus, every irreducible component is of the form $V(\mathfrak{p})$ for some $\mathfrak{p} \in X$.

Now, suppose we have an irreducible component; then, we can write it as $V(\mathfrak{p})$ for some $\mathfrak{p} \in X$. Then, if $\mathfrak{q} \in X$ and $\mathfrak{q} \subset \mathfrak{p}$ properly, then $V(\mathfrak{p}) \subset V(\mathfrak{q})$ properly since $V(\mathfrak{q})$ contains $\mathfrak{q}$ but $V(\mathfrak{p})$ does not contain $\mathfrak{q}$. Since $\mathfrak{q}$ is prime, $V(\mathfrak{q})$ is irreducible, violating the maximality of $V(\mathfrak{p})$. Thus, there is no such $\mathfrak{q}$, and $\mathfrak{p}$ is a minimal prime ideal of A.

Conversely, let $\mathfrak{p}$ be a minimal prime ideal of A. Then, $V(\mathfrak{p})$ is contained in some irreducible component, which we've shown has the form $V(\mathfrak{q})$ for some $\mathfrak{q} \in X$. Thus, $V(\mathfrak{p}) \subset V(\mathfrak{q})$, so $\mathfrak{q} \subset \mathfrak{p}$. Since $\mathfrak{p}$ is a minimal prime ideal, $\mathfrak{p} = \mathfrak{q}$, so $V(\mathfrak{p}) = V(\mathfrak{q})$, and $V(\mathfrak{p})$ is an irreducible component. □

1.21 Exercise

(i) Proof. Let $f \in A$. If $\mathfrak{q} \in Y_{\phi(f)}$, then $\mathfrak{q} \notin V_Y(\phi(f))$, so $(\phi(f)) \not\subset \mathfrak{q}$. Thus, $\phi(f) \notin \mathfrak{q}$, so $f \notin \phi^{-1}(\mathfrak{q})$. Hence, $\phi^{-1}(\mathfrak{q}) \notin V_X(f)$, so $\phi^{-1}(\mathfrak{q}) \in X_f$ and $\mathfrak{q} \in \phi^{*-1}(X_f)$.

Conversely, if $\mathfrak{q} \notin Y_{\phi(f)}$, then $\mathfrak{q} \in V_Y(\phi(f))$, so $\phi(f) \in \mathfrak{q}$. Thus, $f \in \phi^{-1}(\mathfrak{q})$, so $\phi^{-1}(\mathfrak{q}) \in V_X(f)$. Hence, $\phi^{-1}(\mathfrak{q}) \notin X_f$. Then, by definition, $\mathfrak{q} \notin \phi^{*-1}(X_f)$. Thus, $\phi^{*-1}(X_f) = Y_{\phi(f)}$ for any $f \in A$.

Let U be an open set of X, so $U = \bigcup_{f \in I} X_f$. Then,

$$\phi^{*-1}(U) = \phi^{*-1}\left(\bigcup_{f \in I} X_f\right) = \bigcup_{f \in I} \phi^{*-1}(X_f) = \bigcup_{f \in I} Y_{\phi(f)},$$

which is an open set of Y. Thus, $\phi^* : Y \to X$ is continuous. □

(ii) Proof. If $\mathfrak{q} \in \phi^{*-1}(V_X(\mathfrak{a}))$, then $\phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}) \in V_X(\mathfrak{a})$. Hence, $\mathfrak{a} \subset \phi^{-1}(\mathfrak{q})$, so $\phi(\mathfrak{a}) \subset \mathfrak{q}$, so $\mathfrak{q} \in V_Y(\phi(\mathfrak{a})) = V_Y(\mathfrak{a}^e)$.

Conversely, if $\mathfrak{q} \in V_Y(\mathfrak{a}^e)$, then $\phi(\mathfrak{a}) \subset \mathfrak{a}^e \subset \mathfrak{q}$, so $\mathfrak{a} \subset \phi^{-1}(\mathfrak{q})$, hence $\phi^{-1}(\mathfrak{q}) \in V_X(\mathfrak{a})$. Thus, $\mathfrak{q} \in \phi^{*-1}(V_X(\mathfrak{a}))$. Combining both directions, we have $\phi^{*-1}(V_X(\mathfrak{a})) = V_Y(\mathfrak{a}^e)$. □

(iii) Proof. Note that $\phi^*(V(\mathfrak{b})) = \{\mathfrak{q}^c : \mathfrak{b} \subset \mathfrak{q}\}$. So, $\mathfrak{a}$ must contain all of these $\mathfrak{q}^c$. Since all closed sets containing $\phi^*(V(\mathfrak{b}))$ must contain each $\mathfrak{q}^c$, all must contain $\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c$. Since this is itself an ideal, it gives the closure:

$$\overline{\phi^*(V(\mathfrak{b}))} = V\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c\right).$$

Note that

$$\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c = \bigcap_{\mathfrak{b} \subset \mathfrak{q}} \phi^{-1}(\mathfrak{q}) = \phi^{-1}\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}\right) = \phi^{-1}(r(\mathfrak{b})) = r(\mathfrak{b})^c = r(\mathfrak{b}^c).$$

Thus,

$$\overline{\phi^*(V(\mathfrak{b}))} = V\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c\right) = V(r(\mathfrak{b}^c)) = V(\mathfrak{b}^c).$$

□

(iv) Proof. Since ϕ is surjective, we have a ring isomorphism $A/(\ker\phi) \to B$. Thus, we have an inverse homomorphism $\psi : B \to A/(\ker\phi)$. From these maps, we obtain $\psi^* : \operatorname{Spec}(A/(\ker\phi)) \to \operatorname{Spec}(B)$ and $\phi^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A/(\ker\phi))$, which are inverse maps describing a bijection; by (i), both are continuous, so they describe a homeomorphism.

Now, we have to show that the map $\operatorname{Spec}(A/(\ker\phi)) \to V(\ker\phi)$, viewed as a closed subspace of $\operatorname{Spec}(A)$, is a homeomorphism.

But this follows easily from Proposition 1.1, considering the map $A \to A/(\ker\phi)$. More concretely, there is a one-to-one, order-preserving, primality-preserving correspondence between ideals of A containing $\ker\phi$ and ideals of $A/(\ker\phi)$. Thus, the map induced by projection $\operatorname{Spec}(A/\ker\phi) \to V(\ker\phi)$ is a bijection. Because the map preserves the containment relation, it is a closed map ($V(\mathfrak{a}) \to V(\mathfrak{a}^c)$). Since it is a bicontinuous bijection, it is a homeomorphism. □

Remark. *This shows that* $\operatorname{Spec}(A)$ *and* $\operatorname{Spec}(A/\mathfrak{N}_A)$ *are naturally homeomorphic.*

(v) Proof. Suppose $\ker\phi \subset \mathfrak{N}$. Since $Y = V(0)$, by (iii), we have

$$\overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(\phi^{-1}(0)).$$

Since $\ker\phi$ is in the nilradical, which is contained by every prime ideal, $V(\phi^{-1}(0)) = X$, so $\phi^*(Y)$ is dense in X.

Conversely, if $\phi^*(Y)$ is dense in X, then we have, by (iii),

$$X = \overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(\phi^{-1}(0)),$$

so $\phi^{-1}(0) \subset \mathfrak{p}$ for all prime ideals, so $\ker\phi \subset \mathfrak{N}$. □

(vi) Proof. We have two ring homomorphisms $\phi : A \to B$ and $\psi : B \to C$. If $\mathfrak{r} \in \operatorname{Spec}(C)$, then we have a map $(\psi \circ \phi)^* : \operatorname{Spec}(C) \to \operatorname{Spec}(A)$. This takes $\mathfrak{r} \in \operatorname{Spec}(C)$ to $(\psi \circ \phi)^{-1}(\mathfrak{r}) \in \operatorname{Spec}(A)$. But, $(\psi \circ \phi)^{-1} = \phi^{-1} \circ \psi^{-1}$. In particular, $(\psi \circ \phi)^*$ decomposes into maps

$$\operatorname{Spec}(C) \xrightarrow{\psi^*} \operatorname{Spec}(B) \xrightarrow{\phi^*} \operatorname{Spec}(A),$$

so $(\psi \circ \phi)^* = \phi^* \circ \psi^*$. □

(vii) Proof. A is an integral domain with one non-zero prime ideal $\mathfrak{p}$, K is the field of fractions of A, and $B = (A/\mathfrak{p}) \times K$. We define $\phi : A \to B$ by $\phi(x) = (\overline{x}, x)$, where $\overline{x}$ is the image of x in $A/\mathfrak{p}$.

Since ϕ is a ring homomorphism, we have $\phi^* : \operatorname{Spec} B \to \operatorname{Spec} A$. A has only one non-zero prime ideal $\mathfrak{p}$, so $\operatorname{Spec} A = \{0, \mathfrak{p}\}$. For B, the only prime ideal of K is 0, and the only prime ideal of $A/\mathfrak{p}$ is 0. Thus, $\operatorname{Spec} B = \{0 \times K, (A/\mathfrak{p}) \times 0\}$. Note that 0 is not prime in B since $0 = (0 \times K)((A/\mathfrak{p}) \times 0)$. We have $\phi^{-1}(0 \times K) = \mathfrak{p}$ and $\phi^{-1}((A/\mathfrak{p}) \times 0) = 0$. Thus, ϕ^* is bijective.

However, we will show that it is not a homeomorphism. We only need to look at the image $\phi^*(\{(A/\mathfrak{p}) \times 0\}) = \{0\}$. Note that $\{(A/\mathfrak{p}) \times 0\} = V((A/\mathfrak{p}) \times 0)$, so this set is closed. However, $\{0\}$ is not closed since for any $\mathfrak{a}$, $0 \in V(\mathfrak{a})$ implies $\mathfrak{a} \subset 0 \subset \mathfrak{p}$, so $\mathfrak{p} \in \overline{\{0\}}$, i.e. ϕ^* maps a closed set to a non-closed set. Therefore, ϕ^* is not a homeomorphism. □

1.22 Exercise

First, we show that if $A = \prod_{i=1}^n$, then $\bigsqcup_{i=1}^n \operatorname{Spec}(A_j) \cong \operatorname{Spec} A$ as topological spaces.

Proof. Let $A = \prod_{i=1}^n$. We claim that all primes take the form

$$A_1 \times \cdots \times A_{j-1} \times \mathfrak{p}_j \times A_{j+1} \times \cdots \times A_n,$$

where $\mathfrak{p}_j \in \operatorname{Spec} A_j$. If $\pi_j : A \to A_j$ is the projection map, this is equivalent to saying all primes are of the form $\pi_j^{-1}(\mathfrak{p}_j)$ for some j and $\mathfrak{p}_j \in \operatorname{Spec}(A_j)$. First of all, it's clear that ideals of this form are prime since for $a, b \in A$ with $\pi_j(ab) \in \mathfrak{p}$, we have $\pi_j(a)\pi_j(b) \in \mathfrak{p}$, so either $\pi_j(a) \in \mathfrak{p}$ or $\pi_j(b) \in \mathfrak{p}$. Thus, either $a \in \pi_j^{-1}(\mathfrak{p})$ or $b \in \pi_j^{-1}(\mathfrak{p})$, so $\pi_j^{-1}(\mathfrak{p}) \in \operatorname{Spec}(A)$. In particular, we've shown

$$\coprod_{i=1}^n \operatorname{Spec}(A_j) \hookrightarrow \operatorname{Spec} A.$$

Conversely, suppose $\mathfrak{p}$ is a prime of A. Suppose $\pi_i(\mathfrak{p}) \neq A_i$ and $\pi_j(\mathfrak{p}) \neq A_j$ for distinct i, j. Then, let the components of $a \in A$ be all zeros except for the ith slot, where we choose $a_i \in A_i - \pi_i(\mathfrak{p})$. Similarly, we define $b \in A$ by letting the components of b be all zeros except for the jth slot, where we choose $b_j \in A_j - \pi_j(\mathfrak{p})$. Then, neither a nor b is in $\mathfrak{p}$, yet $ab = 0 \in \mathfrak{p}$, violating the primality of $\mathfrak{p}$. Thus, all projection maps of $\mathfrak{p}$ must be surjective except for one. So,

$$\mathfrak{p} = A_1 \times \cdots \times A_{j-1} \times \pi_j(\mathfrak{p}) \times A_{j+1} \times \cdots \times A_n.$$

Let $a_j, b_j \in A_j$. Lift a_j, b_j to $a, b \in A$ by letting all other components be zero. If $ab \in \mathfrak{p}$, then $a_j b_j \in \pi_j(\mathfrak{p})$. Since $\mathfrak{p}$ is prime, either a or b is in $\mathfrak{p}$. Without loss of generality, suppose $a \in \mathfrak{p}$. Then, $\pi_j(a) = a_j \in \pi_j(\mathfrak{p})$. Thus, $\pi_j(\mathfrak{p})$ is prime. Hence, we have a bijection

$$\coprod_{i=1}^n \operatorname{Spec}(A_j) \cong \operatorname{Spec} A.$$

To be more concrete, consider the projection $\pi_j : A \to A_j$. By Exercise 1.21(iv), π_j^* is a homeomorphism of $\operatorname{Spec}(A_j)$ onto the closed subset $V(\ker \pi_j)$ of $\operatorname{Spec} A$. Explicitly, the kernel is $A_1 \times \cdots \times A_{j-1} \times 0 \times A_{j+1} \times \cdots \times A_n$, which matches our constructed maps. We identified $V(A_1 \times \cdots \times A_{j-1} \times 0 \times A_{j+1} \times \cdots \times A_n)$ directly with $\operatorname{Spec} A_j$, which are homeomorphic. To be slightly pedantic, note that the closed subsets just mentioned are disjoint for distinct j by our earlier proof that a prime cannot have two non-surjective projections. □

Proof. By the previous proof, (ii) implies (i) since $A \cong A_1 \times A_2$ implies $\operatorname{Spec}(A) \cong \operatorname{Spec}(A_1) \sqcup \operatorname{Spec}(A_2)$, so $\operatorname{Spec}(A)$ is disconnected.

Now, we show (i) implies (iii). Suppose $\operatorname{Spec}(A)$ is disconnected. By Exercise 1.21(iv), $\operatorname{Spec}(A)$ is naturally homeomorphic to $\operatorname{Spec}(A/\mathfrak{N})$, so consider $A' = A/\mathfrak{N}$, which has no nilradical and is disconnected (by the homeomorphism). We can write $\operatorname{Spec}(A')$ as the disjoint union of two closed sets $V(\mathfrak{a})$ and $V(\mathfrak{b})$ for ideals $\mathfrak{a}, \mathfrak{b} \subset A'$. By Exercise 1.15(iv),

$$\operatorname{Spec}(A') = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}),$$

so $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{N}_{A'} = 0$. Further, since $V(\mathfrak{a}) \cap V(\mathfrak{b}) = \emptyset$, $\mathfrak{a} + \mathfrak{b} = A'$. Consider the ring homomorphism $\psi : A' \to A'/\mathfrak{a} \times A'/\mathfrak{b}$. If $\psi(x) = (0, 0)$, then $x \in \mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{N}_{A'} = 0$, so ψ is injective. By Proposition 1.10(ii), since $\mathfrak{a} + \mathfrak{b} = (1)$, ψ is surjective, so we have a ring isomorphism. Thus,

$$A/\mathfrak{N} \cong A_1 \times A_2$$

for nonzero rings A_1 and A_2. Now, consider the idempotent $\overline{e} = (1, 0)$. Then, there is an $e \in A$ such that $e^2 - e \in \mathfrak{N}$. If $e^2 - e = 0$, then we have our idempotent; otherwise, there is a minimal $n > 0$ such that $(e^2 - e)^n = 0$. We can use the identity in $\mathbb{Z}[X]$ (the binomial theorem), which expands $1 = (X + (1 - X))^{2n-1}$ as

$$1 = \sum_{k=0}^{2n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k}.$$

Let $f = \sum_{k=0}^{n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k}$. Then, we have

$$f = f^2 + f \sum_{k=n}^{2n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k},$$

but every term of f contains $(1 - e)$ of degree at least n, and all terms of the remainder contain e of degree at least n. Thus, the latter sum is annihilated, and we have $f = f^2$. Note that since $e(1-e) \in \mathfrak{N}$, $f \equiv (1-e)^{2n-1} \mod \mathfrak{N}$. Projecting down to $A_1 \times A_2$, we have $\overline{f} = (1-(1,0))^{2n-1} = (0,1)^{2n-1} = (0,1)$, which is not the image of either 0 or 1. Thus, as desired, f is a non-trival idempotent.

Now, we show (iii) implies (ii). Suppose e is an idempotent of A not equal to 0 or 1. Note that

$$(1-e)^2 = 1 - 2e + e^2 = 1 - 2e + e = 1 - e,$$

so $1 - e$ is also an idempotent. Further, note that $e(1 - e) = e - e^2 = e - e = 0$, so e and $1 - e$ are orthogonal idempotents. For any $x \in A$, we have $x = xe + x(1 - e)$. Then, we have the decomposition $\psi : A \cong Ae \times A(1 - e)$ by multiplying by e and $1 - e$, respectively, i.e. $\psi(x) = (xe, x(1 - e))$. Note that the right-hand side is a ring since e acts as the identity in Ae and $(1 - e)$ acts as the identity in $A(1 - e)$. Further, it's easily checked that the map is a ring homomorphism by using the idempotent relation and the orthogonality of the two idempotents. The only thing left to check is that the map is bijective. If $\psi(x) = (0, 0)$, then $xe = 0$ and $x(1 - e) = 0$, so their sum $xe + x(1 - e) = x = 0$, so injectivity holds. Conversely, if we have $(ae, b(1 - e))$ in the direct product, then $ae + b(1 - e) \in A$ maps to this element, so surjectivity holds. Thus, this is an isomorphism of rings. Finally, (iii) implies (ii), and now we see that all three properties are equivalent. □

Remark. *By Exercise 1.12, no local ring has an idempotent distinct from* 0 *and* 1*. We've shown this cannot happen if* $\mathrm{Spec}(A)$ *is disconnected, so the spectrum of a local ring is always connected.*

1.23 Exercise

(i) Proof. For any $f \in A$, consider X_f. Clearly, X_f is closed and open if and only if $V(f)$ is open and closed. Since $V(f)$ is closed by definition, we just have to show that $V(f)$ is an open set, i.e. there is some $g \in A$ such that $V(f) = V(g)^c$. Since A is a boolean ring, $f = f^2$ and $(1-f)^2 = 1-f$. Note that for any $x \in A$, we can write $x = xf + x(1-f)$. Consider the map $\psi : A \to Af \times A(1-f)$. We will show that this is a ring isomorphism. It's a homomorphism since Af and $A(1 - f)$ are rings with multiplicative identity f and $1 - f$; also, since $f(1 - f) = f - f^2 = 0$, multiplication is respected, i.e. $(af+b(1-f))(cf+d(1-f)) = (ac)f+(bd)(1-f)$. Now, we must show it is bijective. If $\psi(x) = (0, 0)$, then $xf = 0$ and $x(1 - f) = 0$, so $0 = xf + x(1 - f) = x$, hence ψ is injective.

Conversely, if we have $(af, b(1-f)) \in Af \times A(1-f)$, then $\psi(af + b(1-f)) = (af, b(1-f))$. Thus, ψ is surjective and hence bijective.

Since ψ realizes A as a direct product of rings, by Exercise 1.22, we know that

$$\operatorname{Spec}(A) \cong \operatorname{Spec}(Af) \sqcup \operatorname{Spec}(A(1-f)).$$

Now, considering the projections $\pi_f : A \to Af$ and $\pi_{1-f}A \to A(1-f)$, we have by Exercise 1.21(iv) that $\operatorname{Spec}(Af) \cong V(f)$ and $\operatorname{Spec}(A(1-f)) \cong V(1-f)$, where $\cong$ denotes homeomorphism. Thus, $\operatorname{Spec}(A)$ is a disjoint union of the closed sets $V(f)$ and $V(1-f)$, so as desired, $V(f) = V(1-f)^c$. □

(ii) Proof. Let $f_1, \ldots, f_n \in A$. We want to show that $X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ for some $f \in A$. Taking complements, this is equivalent to the statement

$$V(f_1) \cap \cdots \cap V(f_n) = V(f).$$

Note that by Exercise 1.15(iii), we have

$$V(f_1) \cap \cdots \cap V(f_n) = V\left(\bigcup_{i=1}^{n}(f_i)\right) = V((f_1, \ldots, f_n)).$$

However, by Exercise 1.11(i), every finitely generated ideal of A is principal, so $(f_1, \ldots, f_n) = (f)$ for some $f \in A$, as desired. □

(iii) Proof. Let $Y \subset X$ be a closed and open set. Since Y is closed and X is quasi-compact by Exercise 1.17(v), we can easily show that Y is also quasi-compact. Let $\{U\}_U$ be an open cover of Y. Then, Y^c is open, so $\{Y^c\} \cup \{U\}_U$ is an open cover of X. Since X is quasi-compact, there is a finite subcover $\{V\}_V$, which obviously covers Y as well. Removing Y^c if necessary, which does not cover any points of Y, we have a finite subcover of Y that uses only open sets in $\{U\}_U$. Thus, Y is quasi-compact.

Since Y is also open, it is a union of basic open sets, and since it is quasi-compact, it is a union of finitely-many basic open sets. In other words, $Y = \bigcup_{i=1}^{n} X_{f_i}$. By (ii), there is some $f \in A$ such that $\bigcup_{i=1}^{n} X_{f_i} = X_f$, so $Y = X_f$, as desired. □

(iv) Proof. Let $\mathfrak{p}, \mathfrak{q}$ be distinct primes in $\operatorname{Spec} A$. Since they are distinct, suppose without loss of generality that $\mathfrak{p}$ contains some $f \in A$ that is not in $\mathfrak{q}$; otherwise, swap $\mathfrak{q}$ and $\mathfrak{p}$. Then, $(f) \subset \mathfrak{p}$, so $\mathfrak{p} \notin X_f$. But, since $\mathfrak{q} \notin V(f)$, we have $\mathfrak{q} \in X_f$. Since X_f is both open and closed by (i), we have two disjoint open sets X_f and X_f^c that contain $\mathfrak{q}$ and $\mathfrak{p}$, respectively. Thus, $\operatorname{Spec} A$ is Hausdorff. Since X is quasi-compact by Exercise 1.17(v), X is a compact Hausdorff space. □

1.24 Exercise

Proof. We start by showing $A(L)$ is a ring. We'll first prove the useful fact:

$$(a \wedge b)' = a' \vee b'.$$

Since the complement is unique, we just have to verify that it is the complement of $a \wedge b$. We have

$$\begin{aligned}(a \wedge b) \wedge (a' \vee b') &= a \wedge b \wedge (a' \vee b') \\ &= a \wedge ((b \wedge a') \vee (b \wedge b')) \\ &= a \wedge ((b \wedge a') \vee 0) \\ &= a \wedge (b \wedge a') \\ &= (a \wedge a') \wedge b \\ &= 0 \wedge b \\ &= 0.\end{aligned}$$

Then, we have

$$\begin{aligned}(a \wedge b) \vee (a' \vee b') &= (a \wedge b) \vee a' \vee b' \\ &= ((a \vee a') \wedge (b \vee a')) \vee b' \\ &= (1 \wedge (b \vee a')) \vee b' \\ &= (b \vee a') \vee b' \\ &= a' \vee (b \vee b') \\ &= a' \vee 1 \\ &= 1,\end{aligned}$$

so indeed, $(a \wedge b)' = a' \vee b'$. In fact, we can permute aa' and $b \mapsto b'$ and take the complement of both sides to obtain the complementary property:

$$a' \wedge b' = (a \vee b)'.$$

Now, we can begin proving that $A(L)$ is a ring, starting with the additive properties. Note that because the complements are unique, we have $0' = 1$ since $0 \wedge 1 = 0$ and $0 \vee 1 = 1$. Also note that since a and a' are symmetric in the definition of a', we have $(a')' = a$.

By the commutativity of $\vee$ and $\wedge$, $+$ is also commutative. We'll now show associativity.

$$\begin{aligned}(a + b) + c &= ((a \wedge b') \vee (a' \wedge b)) + c \\ &= (((a \wedge b') \vee (a' \wedge b)) \wedge c') \vee (((a \wedge b') \vee (a' \wedge b))' \wedge c) \\ &= ((a \wedge b' \wedge c') \vee (a' \wedge b \wedge c')) \vee (((a \wedge b') \vee (a' \wedge b))' \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (((a \wedge b')' \wedge (a' \wedge b)') \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee ((a' \vee b) \wedge (a \vee b') \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (a' \wedge a \wedge c) \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee (b \wedge b' \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (0 \wedge c) \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee (0 \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee 0 \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee 0 \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) \vee (a \wedge b \wedge c).\end{aligned}$$

Note that the final line is completely symmetric under permutations of a, b, and c. Since we already have commutativity, this is equal to $c + (b + a)$. Since permuting a and c does not change the value of the final line (by aforementioned symmetry), we have $(a + b) + c = a + (b + c)$, so addition is associative.

For any $a \in A(L)$, we have

$$a + 0 = (a \wedge 0') \vee (a' \wedge 0) = (a \wedge 1) \vee (a' \wedge 0) = a \vee 0 = a,$$

so 0 is the additive identity. Further, for any $a \in A(L)$, we have

$$a + a' = (a \wedge a') \vee (a \wedge a') = a \wedge a' = 0,$$

by definition, so each a has an additive inverse. Thus, $(A(L), +)$ forms an abelian group.

Moving on, note that multiplication is commutative because $\wedge$ is commutative. Then, let $a, b \in A(L)$. We have $a \cdot 1 = a \wedge 1 = a$ since 1 is a global max. Thus, $A(L)$ has commutative multiplication with unit. Associativity follows easily since

$$a \cdot (b \cdot c) = a \wedge (b \wedge c) = (a \wedge b) \wedge c = (a \cdot b) \cdot c.$$

Lastly, we must show that multiplication distributes over addition. We have

$$\begin{aligned}
ab + ac &= (a \wedge b) + (a \wedge c) \\
&= ((a \wedge b) \wedge (a \wedge c)') \vee ((a \wedge b)' \wedge (a \wedge c)) \\
&= (a \wedge b \wedge (a' \vee c')) \vee ((a' \vee b') \wedge a \wedge c) \\
&= (a \wedge b \wedge a') \vee (a \wedge b \wedge c') \vee (a' \wedge a \wedge c) \vee (b' \wedge a \wedge c) \\
&= (0 \wedge b) \vee (a \wedge b \wedge c') \vee (0 \wedge c) \vee (b' \wedge a \wedge c) \\
&= 0 \vee (a \wedge b \wedge c') \vee 0 \vee (b' \wedge a \wedge c) \\
&= (a \wedge b \wedge c') \vee (a \wedge b' \wedge c) \\
&= a \wedge ((b \wedge c') \vee (b' \wedge c)) \\
&= a \wedge (b + c) \\
&= a(b + c),
\end{aligned}$$

so the distributive property holds, and $A(L)$ is a ring. It is now easy to see that it is a Boolean ring since for any $a \in A(L)$, $a^2 = a \wedge a = a$.

Conversely, suppose A is a Boolean ring. We define an ordering $a \leq b$ when $a = ab$. Thus, 0 is clearly a minimal element and 1 is clearly a maximal element, satisfying property (i) of a Boolean lattice. Define $a \wedge b = ab$ and $a \vee b = a + b + ab$. We now verify the two distributive properties:

$$\begin{aligned}
a \wedge (b \vee c) &= a(b + c + bc) \\
&= ab + ac + abc \\
&= ab + ac + a^2bc \\
&= ab + ac + (ab)(ac) \\
&= (ab) \vee (ac) \\
&= (a \wedge b) \vee (a \wedge c).
\end{aligned}$$

Also,

$$\begin{aligned}(a \vee b) \wedge (a \vee c) &= (a + b + ab)(a + c + ac) \\ &= a^2 + ac + a^2c + ab + bc + abc + a^2b + abc + a^2bc \\ &= a + ac + ac + ab + bc + abc + ab + abc + abc \\ &= a + 2ac + 2ab + bc + abc + 2abc \\ &= a + bc + abc \\ &= a \vee (bc) \\ &= a \vee (b \wedge c),\end{aligned}$$

so property (ii) is satisfied.

Lastly, we must show that there is a unique complement a' for each $a \in A$. We claim that this is satisfied by $1 - a$. We can prove existence and uniqueness at once by applying the definitions. We have

$$0 = a \wedge a' = aa' \quad \text{and} \quad 1 = a \vee a' = a + a' + aa'.$$

Since $aa' = 0$, we have that $1 = a + a' + 0$, so $a' = 1 - a$. This proves (iii). Thus, A defined by our $\wedge$ and $\vee$ is a Boolean lattice. □

1.25 Exercise

Proof. Here, we prove Stone's theorem. Let L be a Boolean lattice. By Exercise 1.24, L can be made into a Boolean ring, $A(L)$. We define $\mathcal{T}$ to be the open-and-closed subsets of $\mathrm{Spec}(A(L))$. By Exercise 1.23, the elements of $\mathcal{T}$ are exactly X_f, where $f \in A(L)$. By Exercise 1.23, $\mathrm{Spec}(A(L))$ is a compact Hausdorff space, so $\mathcal{T}$ is the set of open-and-closed subsets of a compact Hausdorff space. Thus, we aim to show that $L \to \mathcal{T}$ is an isomorphism of lattices.

We show that $\mathcal{T}$ is a Boolean lattice. We define $U \leq V$ when $U \subset V$. (i) X_1 is the maximal element; X_0 is the minimal element. (ii) We define $\wedge$ and $\vee$ by $\cap$ and $\cup$. These are closed operations (two open-and-closed sets yield an open-and-closed set), and they distribute over each other. (iii) We define the complement of X_f by $X_f^c = X_{f'}$; there is such an f' since X_f^c is open-and-closed. $X_f \wedge X_{f'} = X_f \cap X_{f'} = \emptyset = X_0$ and $X_f \vee X_{f'} = X_f \cup X_{f'} = X = X_1$. Thus, $\mathcal{T}$ is a lattice.

The map in view factors as $L \to A(L) \to \mathcal{T}$, where the first map is clearly a bijection. Suppose $f, g \in A(L)$. Then, if $F, G \in L$ yield $f, g \in A(L)$, then $fg = F \wedge G$ by our construction of the Boolean ring (Exercise 1.24). Thus, $F \wedge G \mapsto fg \mapsto X_{fg}$. However, $F \mapsto f \mapsto X_f$ and $G \mapsto g \mapsto X_g$. However, $X_f \wedge X_g = X_f \cap X_g = X_{fg}$ by Exercise 1.17, so the map preserves $\wedge$. If f, f' are complements in L, then in $A(L)$, they satisfy

$$f + f' = (f \wedge f) \vee (f' \wedge f') = f \vee f' = 1,$$

so $f' = 1 - f$. However, $(f) + (1 - f) = (1)$ and $(f) \cap (1 - f)$ contains no prime ideals. Thus, $X_f^c = X_{1-f}$, so the map preserves complements. Because the map preserves complements and $\wedge$, de Morgan's law here shows that it also preserves $\vee$. In particular, $L \to \mathcal{T}$ is a homomorphism of lattices.

Lastly, we show bijectivity. Note that $L \to A(L)$ is a bijection since the map preserves the underlying set. Every element of $\mathcal{T}$ has the form X_f for some $f \in A$, and the map is $f \mapsto X_f$, so clearly the map is surjective. To show injectivity, suppose $f, g \in X_f$ and $X_f = X_g$. Then, for every prime ideal $\mathfrak{p}$ of $A(L)$, $f \in \mathfrak{p} \iff g \in \mathfrak{p}$. By Exercise 1.11, $A/\mathfrak{p}$ is the field of two

elements, so for all $\mathfrak{p}$, $f \equiv g \mod \mathfrak{p}$. In particular, $f - g \in \mathfrak{p}$ for all $\mathfrak{p}$. Hence, $f - g \in \mathfrak{N}$. Since $a^2 = a$ for all $a \in A(L)$ (Boolean ring), $(f-g)^2 = f - g$. However, $f - g$ is nilpotent, so $(f-g)^n = 0$ for some $n > 0$. By a simple inductive argument using $(f-g)^2 = f - g$, we have $f - g = (f-g)^n = 0$, so $f = g$. Thus, the map is injective and therefore bijective. Hence, $L \to \mathcal{T}$ is a lattice isomorphism. □

1.26 Exercise

Let A be a ring. We define $\mathrm{Max}(A)$ to be the subspace of $\mathrm{Spec}(A)$ containing the maximal ideals, and we endow it with the induced topology. It is called the *maximal spectrum* of A. Let X be a compact Hausdorff space, and let $C(X)$ denote the ring of all real-valued continuous functions on X. For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal $\mathfrak{m}_x$ is maximal since it is the kernel of the map $C(X) \to \mathbb{R}$ via $f \mapsto f(x)$. Thus, we have a map $\mu : X \to \mathrm{Max}(C(X))$ via $x \mapsto \mathfrak{m}_x$. We aim to show that μ is a homeomorphism of X onto $\mathrm{Max}(C(X))$.

Proof. Let $\mathfrak{m} \in \mathrm{Max}(C(X))$, and define $V = V(\mathfrak{m})$ by

$$V = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}.$$

Suppose V is empty. Then, for each $x \in X$, there exists $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighborhood U_x of x in X on which f_x does not vanish. By compactness, a finite number of the neighborhoods, say $U_{x_1}, \ldots, U_{x_n}$ cover X. Let

$$f = f_{x_1}^2 + \cdots + f_{x_n}^2.$$

Then, f does not vanish anywhere on X, hence $1/f$ is a well-defined real-valued function on X. Open intervals are a basis of $\mathbb{R}$, so since 0 is not in the image of $1/f$, it is enough to consider intervals (a, b), where $0 < a < b$ or $a < b < 0$. Note that $(1/f)^{-1}((a,b)) = f^{-1}((1/b, 1/a))$, but this is open by the continuity of f, so $1/f$ is also continuous. Hence, $1/f \in C(X)$ and f is a unit, violating the maximality of $\mathfrak{m}$. Hence, V is non-empty, and there is some $x \in V$. Then, $\mathfrak{m} \subset \mathfrak{m}_x$, so by maximality, $\mathfrak{m} = \mathfrak{m}_x$. Hence, μ is surjective.

Since X is compact and Hausdorff, it is also normal. By the Hausdorff property, with distinct $x, y \in X$, we have disjoint open sets U_x and U_y such that $x \in U_x$ and $y \in U_y$. By normality, there are disjoint closed sets V_x and V_y containing U_x and U_y, respectively. By Urysohn's lemma (since X is normal), there exists a continuous function which is 0 on V_x and 1 on V_y, and vice-versa. Thus, $\mathfrak{m}_x$ contains an element that $\mathfrak{m}_y$ does not and vice-versa, so $\mu : x \mapsto \mathfrak{m}_x$ is injective. Compiling results, we've shown that μ is bijective.

Let $f \in C(X)$. Define

$$U_f = \{x \in X : f(x) \neq 0\} \quad \text{and} \quad \tilde{U}_f = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : f \notin \mathfrak{m}\}.$$

Consider any closed set C of X. If $y \notin C$, then since $\{y\}$ is closed, we have by Urysohn's lemma that there is some $f \in C(X)$ such that $f(y) = 1$ and f is 0 on C. Thus, C is the intersection of U_f^c for all $f \in C(X)$ such that f is zero on C. Thus, any open set is a union of U_f, so $\{U_f\}$ forms a basis of X. Also, for any $f \in C(X)$, X_f forms a basis of $\mathrm{Spec}(C(X))$ by Exercise 1.17. Restricted to $\mathrm{Max}(C(X))$, these open sets have the form

$$\{\mathfrak{m} \in \mathrm{Max}(C(X)) : (f) \subset \mathfrak{m}\}^c = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : (f) \not\subset \mathfrak{m}\} = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : f \notin \mathfrak{m}\} = \tilde{U}_f,$$

so $\{\tilde{U}_f\}$ defines a basis for $\mathrm{Max}(C(X))$ since the topology is inherited from $\mathrm{Spec}(C(X))$.

Thus, if we can show $\mu(U_f) = \tilde{U}_f$, we will have shown that μ is an open map and thus a homeomorphism by its already established bijectivity. If $f(x) \neq 0$, then $f \notin \mathfrak{m}_x$. Conversely, if $f \notin \mathfrak{m}_x$, then by definition, $f(x) \neq 0$. Thus, $\mu(U_f) = \tilde{U}_f$, completing the proof. □

Remark. *This implies the topological space X can be reconstructed from the ring $C(X)$ via* $\mathrm{Max}(C(X))$.

1.27 Exercise

Let k be an algebraically closed field, and let

$$f_a(x_1, \ldots, x_n) = 0$$

be a set of polynomial equations in n variables with coefficients in k. The set X of all points $x = (x_1, \ldots, x_n) \in k^n$ which satisfy the equations is an *affine algebraic variety*. Consider the set of all polynomials $g \in k[x_1, \ldots, x_n]$ with the property that $g(x) = 0$ for all $x \in X$. This set is an ideal $I(X)$ in the polynomial ring, and is called the *ideal of the variety* X. The quotient ring

$$P(X) = k[x_1, \ldots, x_n]/I(X)$$

is the ring of polynomial functions on X because two polynomials g, h define the same polynomial on X if and only if $g - h$ vanishes at every point of X, i.e. if and only if $g - h \in I(X)$.

Let ξ_i be the image of t_i under the projection $k[x_1, \ldots, x_n] \to P(X)$. These are the *coordinate functions* on $P(X)$. If $x \in X$, then $\xi_i(x)$ is well-defined and gives the ith coordinate of x. Clearly, $P(X)$ is generated as a k-algebra by the coordinate functions, and is called the *coordinate ring* (or *affine algebra*) of X.

As in Exercise 1.26, for each $x \in X$, let $\mathfrak{m}_x$ be the ideal of all $f \in P(X)$ such that $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, we have a mapping $\mu : X \to \mathrm{Max}(P(X))$ by $x \mapsto \mathfrak{m}_x$. It is easy to show that μ is injective: if $x \neq y$, we have $\xi_i(x) \neq \xi_i(y)$ for some i, so $\xi_i - \xi_i(x) \in \mathfrak{m}_x$, but it is not in $\mathfrak{m}_y$. Therefore, $\mathfrak{m}_x \neq \mathfrak{m}_y$. We now show that μ is surjective; this creates a one-to-one correspondence between X and $\mathrm{Max}(P(X))$, one form of Hilbert's Nullstellensatz.

We'll start by showing that the maximal ideals of $k[x]$ are exactly $(x - a)$ for $a \in k$.

Proof. Clearly, $(x-a)$ is maximal as it is the kernel of the surjective ring homomorphism $k[x] \to k$ via $f \mapsto f(a)$ since k is algebraically closed. Let $\mathfrak{m}$ be a maximal ideal of $k[x]$. Let $f \in \mathfrak{m}$. Since the degree of a polynomial is a nonnegative integer, there is a minimal degree m of $\mathfrak{m}$ (it's clear that $\mathfrak{m} \neq 0$). Note that $m > 0$; otherwise $\mathfrak{m}$ contains a unit, violating maximality. Let f be a polynomial of degree m. Then, since k is algebraically closed, it factors as $a_0(x-a_1)\cdots(x-a_m)$ for $a_0 \neq 0$. Redefine f as $a_0^{-1}f$. Since $\mathfrak{m}$ is an ideal and k is a subring of $k[x]$, $f = (x-a_1)\cdots(x-a_m)$ is in $\mathfrak{m}$. If g is also a monic degree m polynomial of $\mathfrak{m}$, then if $f - g$ is nonzero, it has degree less than m, violating the minimality of m. Hence, $f = g$, so all degree m polynomials in $\mathfrak{m}$ have $a_1, \ldots, a_m$ as roots.

Let g be a polynomial in $\mathfrak{m}$ of minimal degree such that $g(a_1) \neq 0$. Then, it has degree $n > m$ since f is not proportional to g (values at a_1). Multiply by some element of k to make it monic; it's still in $\mathfrak{m}$. Since k is algebraically closed, $g(x) = (x-b_1)\cdots(x-b_n)$. Now, $f(x)(x-b_{m+1})\cdots(x-b_n)$ is in $\mathfrak{m}$ since $(x-b_{m+1})\cdots(x-b_n) \in k[x]$. Thus,

$$h(x) = f(x)(x-b_{m+1})\cdots(x-b_n) - g(x) = (f(x) - (x-b_1)\cdots(x-b_m))(x-b_{m+1})\cdots(x-b_n).$$

Note that $h(x) \in \mathfrak{m}$. Also, $f(x)-(x-b_1)\cdots(x-b_m) \neq 0$ since $f(a_1) = 0$, but $(x-b_1)\cdots(x-b_m) \neq 0$. However, both are monic of degree m, so their difference has degree less than m. Thus, $h(x)$ has degree less than n. However, $f(x)(x-b_{m+1})\cdots(x-b_n)$ evaluates to 0 at a_1, but $g(a_1) \neq 0$, so $h(a_1) \neq 0$. This violates the minimality of n. Thus, we have that no such g exists, meaning that $g(a_1) = 0$ for all $g \in \mathfrak{m}$. Hence, $\mathfrak{m} \subset (x - a_1)$, so by maximality, $\mathfrak{m} = (x - a_1)$. □

Now, we show that the maximal ideals of $k[x_1, \ldots, x_n]$ are exactly $(x_1 - a_1, \ldots, x_n - a_n)$ for $a_i \in k$.

Proof. Consider the ideal $(x_1 - a_1, \ldots, x_n - a_n)$ of $k[x_1, \ldots, x_n]$ for $a_i \in k$. It is the kernel of the surjective ring homomorphism $k[x_1, \ldots, x_n] \to k$ via $f \mapsto f(a_1, \ldots, a_n)$, so it is a maximal ideal.

Conversely, suppose $\mathfrak{m}$ is a maximal ideal of $k[x_1, \ldots, x_n]$. For each i, we have an injective ring homomorphism $k[x_i] \to k[x_1, \ldots, x_n]$. The contraction of $\mathfrak{m}$ is a prime ideal of $k[x_i]$. This is contained in some maximal ideal $(x_i - a_i)$ by the previous proof. Note that $\mathfrak{m} \subset (x_i - a_i) + \mathfrak{m}$; since the contraction of $\mathfrak{m}$ is contained in $(x_i - a_i)$, the contraction of the sum is also contained in $(x_i - a_i)$. Thus, the sum cannot be the whole ring, so by the maximality of $\mathfrak{m}$, we have $(x_i - a_i) \subset \mathfrak{m}$. Doing this for each i, we have that $(x_i - a_i)$ is a subset of $\mathfrak{m}$ for each i. Thus, $(x_1 - a_1, \ldots, x_n - a_n) \subset \mathfrak{m}$, so by the maximality of both sets, $\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$. □

Now, we can prove the surjectivity of μ.

Proof. Suppose $\mathfrak{m} \in \mathrm{Max}(P(X))$. Using the ring homomorphism $\phi : k[x_1, \ldots, x_n] \to P(X)$, we have that $\phi^{-1}(\mathfrak{m})$ is a prime ideal of $k[x_1, \ldots, x_n]$. Let $\mathfrak{M}$ be a maximal ideal of $k[x_1, \ldots, x_n]$ containing $\phi^{-1}(\mathfrak{m})$. Since ϕ is surjective, $\phi(\mathfrak{M})$ is an ideal of $P(X)$ containing $\mathfrak{m}$. By maximality, $\phi(\mathfrak{M}) = \mathfrak{m}$ or $\phi(\mathfrak{M}) = P(X)$.

In the first case, $\mathfrak{M} \subset \phi^{-1}(\mathfrak{m})$, so $\mathfrak{M} = \phi^{-1}(\mathfrak{m})$, and $\phi^{-1}(\mathfrak{m})$ is a maximal ideal of $k[x_1, \ldots, x_n]$. In the second case, since $P(X) = k[x_1, \ldots, x_n]/I(X)$, we have that $I(X) + \mathfrak{M} = k[x_1, \ldots, x_2]$. In particular, since $\mathfrak{M}$ is maximal, $\mathfrak{M}$ does not contain $I(X)$; else, $k[x_1, \ldots, x_2] = I(X) + \mathfrak{M} = \mathfrak{M}$. But, $I(X) \subset \phi^{-1}(\mathfrak{m})$, violating the fact that $\mathfrak{M}$ contains $\phi^{-1}(\mathfrak{m})$. Hence, we have that $\phi^{-1}(\mathfrak{m})$ is a maximal ideal of $k[x_1, \ldots, x_n]$.

By our previous proof, we then have that $\phi^{-1}(\mathfrak{m}) = (x_1 - a_1, \ldots, x_n - a_n)$ for $a_i \in k$. Since $I(X) \subset \phi^{-1}(\mathfrak{m})$, we must have $a = (a_1, \ldots, a_n) \in X$. By the surjectivity of the projection map ϕ, $\phi((x_1 - a_1, \ldots, x_n - a_n)) = \mathfrak{m}$. Since for any $f \in \phi((x_1 - a_1, \ldots, x_n - a_n))$, $f(a) = 0$, we have that $\mathfrak{m}_a \subset \mathfrak{m}$. By maximality, $\mathfrak{m} = \mathfrak{m}_a$, proving the surjectivity of μ. □

1.28 Exercise

Let $f_1, \ldots, f_m$ be elements of $k[x_1, \ldots, x_n]$. They determine a *polynomial mapping* $\phi : k^n \to k^m$; if $x \in k^n$, the cooordinates of $\phi(x)$ are $f_1(x), \ldots, f_m(x)$. Let X, Y be affine algebraic varieties in k^n, k^m, respectively. A mapping $\phi : X \to Y$ is said to be *regular* if ϕ is the restriction to X of a polynomial mapping from k^n to k^m. If η is a polynomial function on Y, then $\eta \circ \phi$ is a polynomial function on X. Hence ϕ induces a k-algebra homomorphism $P(Y) \to P(X)$, namely $\eta \mapsto \eta \circ \phi$. We will show that this yields a one-to-one correspondence between the regular mappings $X \to Y$ and the k-algebra homomorphisms $P(Y) \to P(X)$.

Proof. Let $\phi : X \to Y$ be a regular mapping. We'll start by showing that the k-homomorphism $\eta \mapsto \eta \circ \phi$ is unique to ϕ. Suppose $\psi : X \to Y$ is a regular function and $\eta \circ \psi = \eta \circ \phi$ for all $\eta \in P(Y)$. Consider the coordinate functions $\xi_i \in P(Y)$. Then, $\xi_i \circ \psi = \xi_i \circ \phi$ for all i, so $\psi = \phi$.

Now, consider an arbitrary k-algebra homomorphism $\theta : P(Y) \to P(X)$, or

$$\theta : k[y_1, \ldots, y_m]/I(Y) \to k[x_1, \ldots, x_n]/I(X).$$

Since ξ_i generate $P(Y)$ as a k-algebra, θ is defined entirely by $\theta(\xi_i)$. Define θ_1 to be the map

$$\theta_1 : k[y_1, \ldots, y_m] \to k[x_1, \ldots, x_n]/I(X)$$

that sends $y_i \mapsto \theta(\xi_i)$. Then, θ_1 is a k-algebra homomorphism. Likewise, there are Ξ_i that generate $P(X)$. Hence, $\theta_1(y_i) = \sum_{\alpha \in I_i} c_{i,\alpha} \Xi^\alpha$ for $c_{i,\alpha} \in k$. Then, we can define ψ to be the unique k-algebra homomorphism

$$\psi : k[y_1, \ldots, y_m] \to k[x_1, \ldots, x_n]$$

defined by $\psi(y_i) = \sum_{\alpha \in I_i} c_{i,\alpha} x^\alpha$. Then, if the canonical projection maps defining the varieties are $\pi_Y : k[y_1, \ldots, y_m] \to k[y_1, \ldots, y_m]/I(Y)$ and $\pi_X : k[x_1, \ldots, x_n] \to k[x_1, \ldots, x_n]/I(X)$, we have $\pi_X \circ \psi = \theta \circ \pi_Y$ by our definition of ψ. In particular, $\pi_X(\psi(I(Y))) = \theta(0) = 0$, so $\psi : I(Y) \to I(X)$.

For each y_i, $\psi(y_i)$ is a polynomial of $k[x_1, \ldots, x_n]$. Hence, $(\psi(y_1)(x), \ldots, \psi(y_m)(x))$ defines a map $k[x_1, \ldots, x_n] \to k[y_1, \ldots, y_m]$. Since $\psi : I(Y) \to I(X)$, this defines a map $X \to Y$. This is regular as it is the restriction of a k-homomorphism $k[x_1, \ldots, x_n] \to k[y_1, \ldots, y_m]$.

The regular functions $X \to Y$ are one-to-one with the k-homomorphisms $P(Y) \to P(X)$. $\square$

2 Modules (In Progress)

2.1 Exercise - Complete

Proof. Let $x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$. Since m, n are coprime, Bezout's lemma says that there are $a, b \in \mathbb{Z}$ such that $am + bn = 1$. Then,

$$x \otimes y = am(x \otimes y) + bn(x \otimes y) = (amx) \otimes y + x \otimes (bny) = 0 \otimes y + x \otimes 0 = 0,$$

so $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$. □

2.2 Exercise - Complete

Proof. Consider the exact sequence

$$0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0.$$

Tensoring with M, we have, by the right exactness of tensor products,

$$\mathfrak{a} \otimes_A M \to A \otimes_A M \to (A/\mathfrak{a}) \otimes_A M \to 0.$$

Under canonical isomorphism, we have $A \otimes_A M = M$. Through this canonical isomorphism the image of $\mathfrak{a} \otimes_A M$ is generated via $a \otimes m \mapsto am \in \mathfrak{a}M$. In fact, any element of $\mathfrak{a}M$ has the form $\sum_i a_i m_i$ for $a_i \in \mathfrak{a}$ and $m_i \in M$; this can be obtained through the map via the element $\sum_i (a_i, m_i) \in \mathfrak{a} \otimes_A M$. By exactness, we have

$$(A/\mathfrak{a}) \otimes_A M \cong (A \otimes_A M)/\operatorname{im}(\mathfrak{a} \otimes_A M) \cong M/\mathfrak{a}M,$$

completing the proof. □

2.3 Exercise - Complete

Proof. Suppose $M \otimes_A N = 0$ for a local ring A and finitely-generated A-modules M, N. If $k = A/\mathfrak{m}$ is the residue field, let $M_k = k \otimes_A M = (A/\mathfrak{m}) \otimes_A M \cong M/\mathfrak{m}M$ (Exercise 2.2). Since $M \otimes_A N = 0$, we have

$$0 = k \otimes_A (M \otimes_A N) \otimes_A k = M_k \otimes_A N_k.$$

By Exercise 2.2, we have

$$M_k \otimes_A N_k \cong (M/\mathfrak{m}M) \otimes_A (N/\mathfrak{m}N).$$

Note that the A-linearity of $(M/\mathfrak{m}M) \otimes_A (N/\mathfrak{m}N)$ factors through $k = A/\mathfrak{m}$, so

$$0 = M_k \otimes_A N_k \cong M_k \otimes_k N_k.$$

However, note that since M and N are finitely-generated, M_k and N_k are finite-dimensional vector spaces over k. Thus, since the tensor product vanishes, either M_k or N_k must vanish (dimensions multiply). By symmetry, assume $M_k = 0$ without loss of generality.

Since A is a local ring, $\mathfrak{m}$ is identical to the Jacobson radical. Since $M/\mathfrak{m}M \cong M_k = 0$, we have that $\mathfrak{m}M = M$. Since M is also finitely-generated, we have by Nakayama's lemma that $M = 0$, as desired. □

2.4 Exercise

2.5 Exercise

2.6 Exercise

2.7 Exercise

2.8 Exercise

2.9 Exercise

2.10 Exercise

2.11 Exercise

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2.27 Exercise

2.28 Exercise

3 Rings and Modules of Fractions

4 Primary Decomposition

5 Integral Dependence and Valuations

6 Chain Conditions

7 Noetherian Rings

8 Artin Rings

9 Discrete Valuation Rings and Dedekind Domains

10 Completions

11 Dimension Theory